LAB

Building an AI agent using watsonx Agent Lab

IBM **SkillsBuild**

Contents

Introduction

In this lab, you'll design and deploy a personalized AI agent using watsonx Agent Lab to support fitness, meal planning, and daily health guidance. This drag-and-drop tool enables you to build and validate a real-world healthcare solution without any coding experience.

Software requirements

To complete this lab, you'll need access to an IBM Cloud account with watsonx.ai and Agent Lab enabled. These platforms provide the tools to build and deploy your AI agent. You'll also need a modern web browser such as Chrome, Firefox, or Edge to access the lab environment smoothly.

Basic familiarity with AI concepts or low-code platforms might help you better understand the process.

Objective

After completing this lab, you should be able to:

- Build an AI agent using IBM watsonx.ai

Lab steps

This lab requires you to complete the following steps:

- **Step 1:** Log in to watsonx.ai
- **Step 2:** Configure the project for Agent Lab
- **Step 3:** Build and configure the Personal Health Advisor agent
- **Step 4:** Deploy and validate the agent using visual tools

Estimated duration to complete

30 minutes

Scenario

Background information

A wellness startup called VitaWell provides digital health solutions to its users, including fitness tracking, workout routines, and meal planning. As its customer base grows, the company is exploring the use of a virtual AI agent to deliver personalized, always-available support tailored to each user's goals and health conditions. You've been hired as the lead AI designer to develop this AI-enabled Personal Health Advisor using watsonx Agent Lab.

Challenge

Currently, the team relies on manual content delivery for health advice through blogs and chat scripts. Health coaches are overworked, which leads to delayed response times, generic guidance, and limited user interaction. The company needs a more interactive, responsive, and personalized experience that adapts to each user's unique lifestyle and goals.

Solution

You'll use watsonx Agent Lab to build a dynamic AI agent powered by the llama-3-3-70B-instruct model. The agent will gather user information and preferences, retrieve contextual data using tools, generate health plans, provide meal suggestions, and curate fitness routines while maintaining a conversational tone and protecting users' private data.

Step 1: Log in to watsonx.ai

Overview

In this step, you'll create or sign in to your IBM Cloud account to access the watsonx.ai environment. IBM Cloud provides the infrastructure and tools needed to build, deploy, and manage AI solutions. You'll develop your AI agent on watsonx.ai, using built-in tools and features that enhance its ability to interpret prompts and respond to people. This setup ensures you have the environment and access needed to build and run the agent powered by the llama-3-3-70B-instruct model.

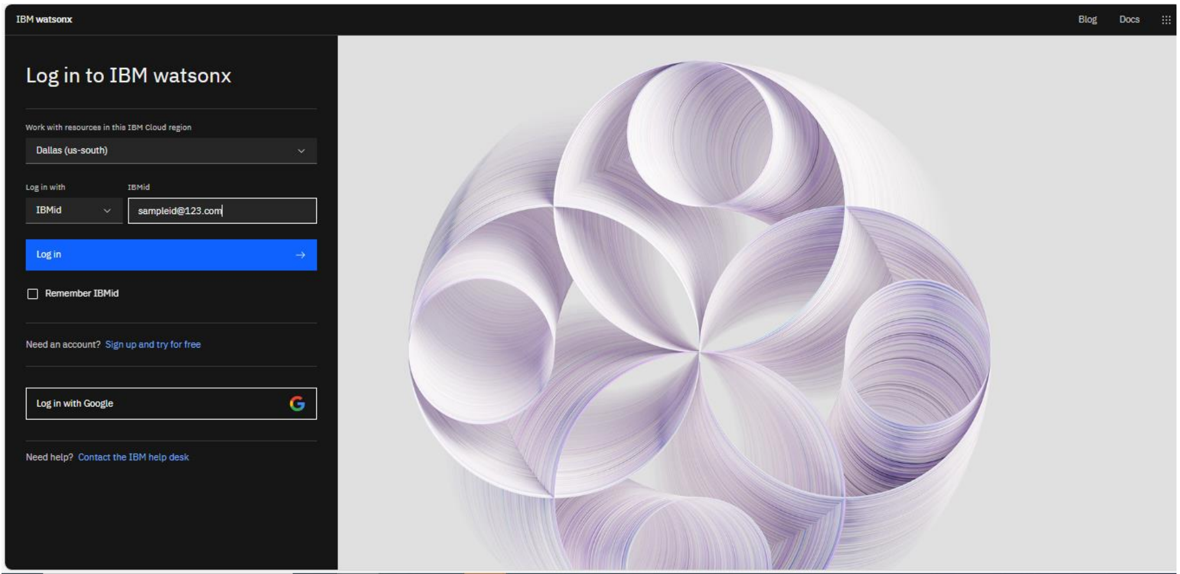
Instructions

1. Go to the [watsonx.ai platform]() and select the **Sign up and try for free** option.

   Graphic note: Edit the screenshot to remove the "URL section".

Alt text: An image showing the watsonx.ai welcome screen with the **Sign up and try for free** option

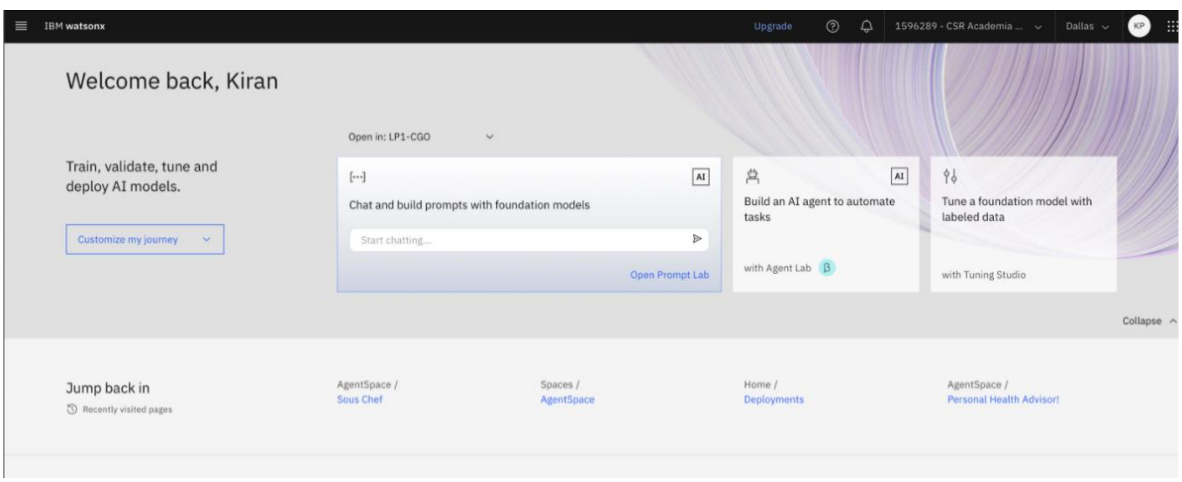
2. Enter your **email** address, create a **password**, and fill in other account details. Then, select **Create account** or **Log in**.



Alt text: An image showing the watsonx.ai welcome screen with the **Log in** button, along with the **Sign up and try for free** option

3. After you've signed in, you'll be able to view the "watsonx.ai" home page.

Graphic note: Global: please remove any name references from the screenshot.



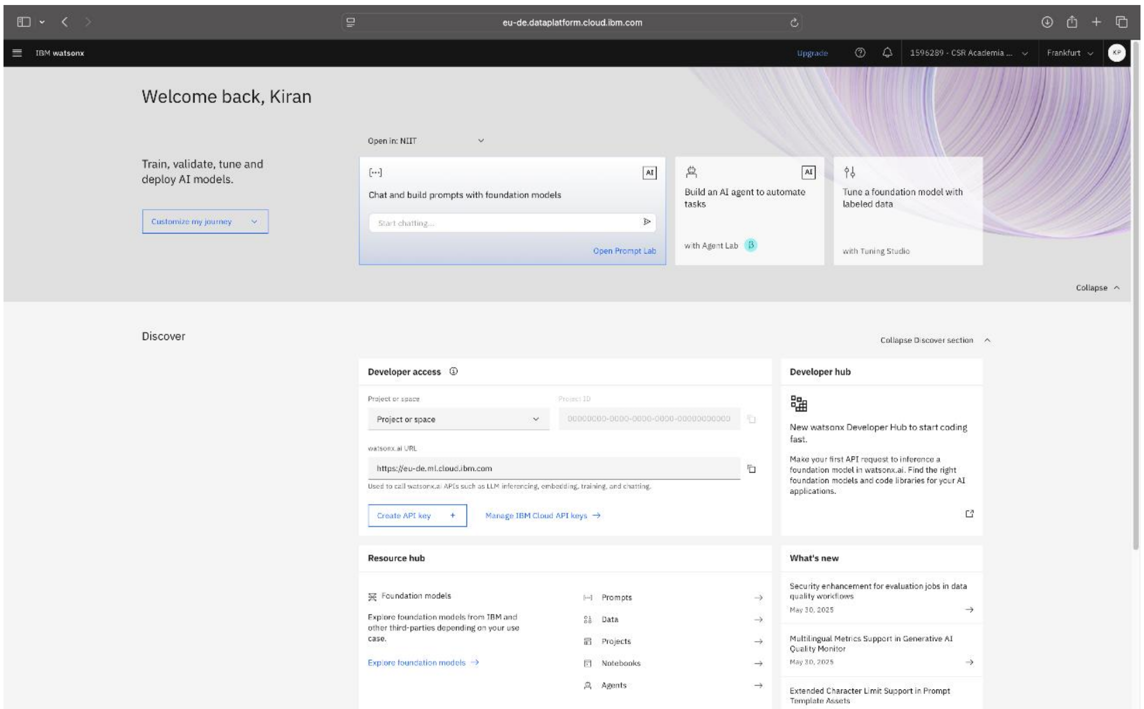
Alt text: An image showing the "watsonx.ai" home page

Step 2: Configure the project for the Agent Lab environment

Overview
In this step, you'll set up your project environment and choose the runtime that will execute your AI agent in watsonx Agent Lab. This setup is essential to ensure that the agent runs smoothly and has access to the right tools and resources.

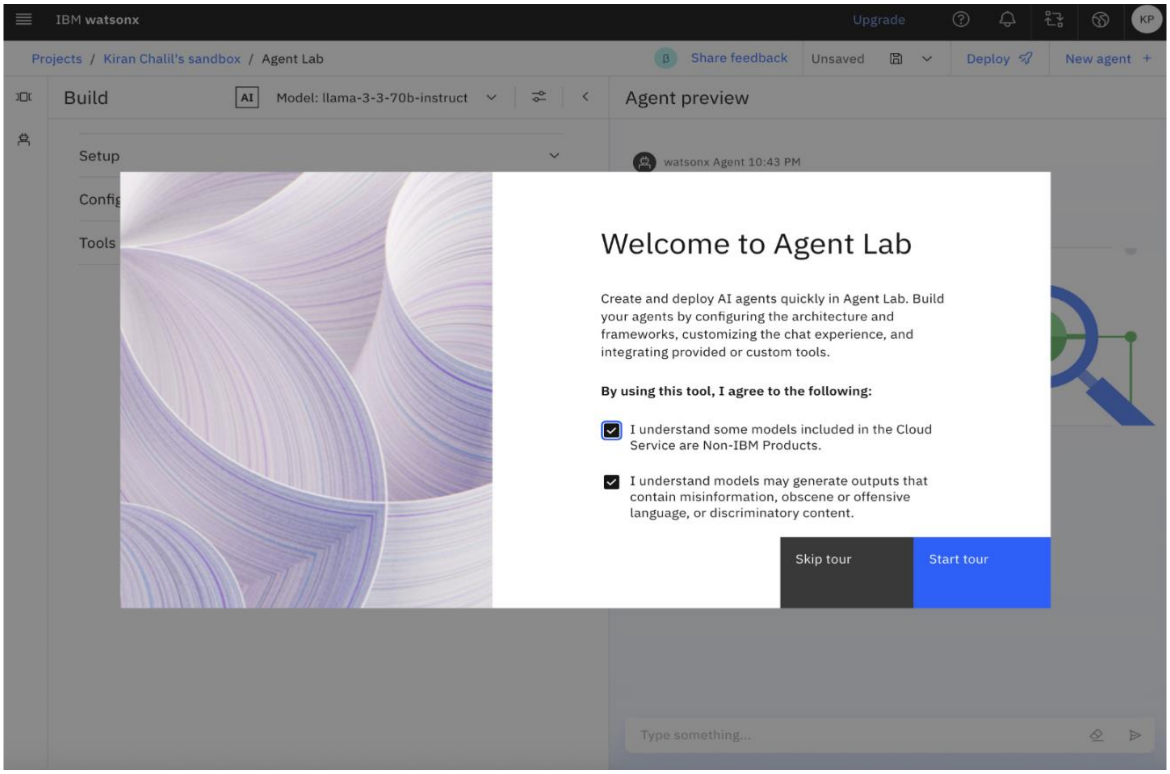
Instructions

1. On the "watsonx.ai" home page, select **Build an AI agent to automate tasks**.



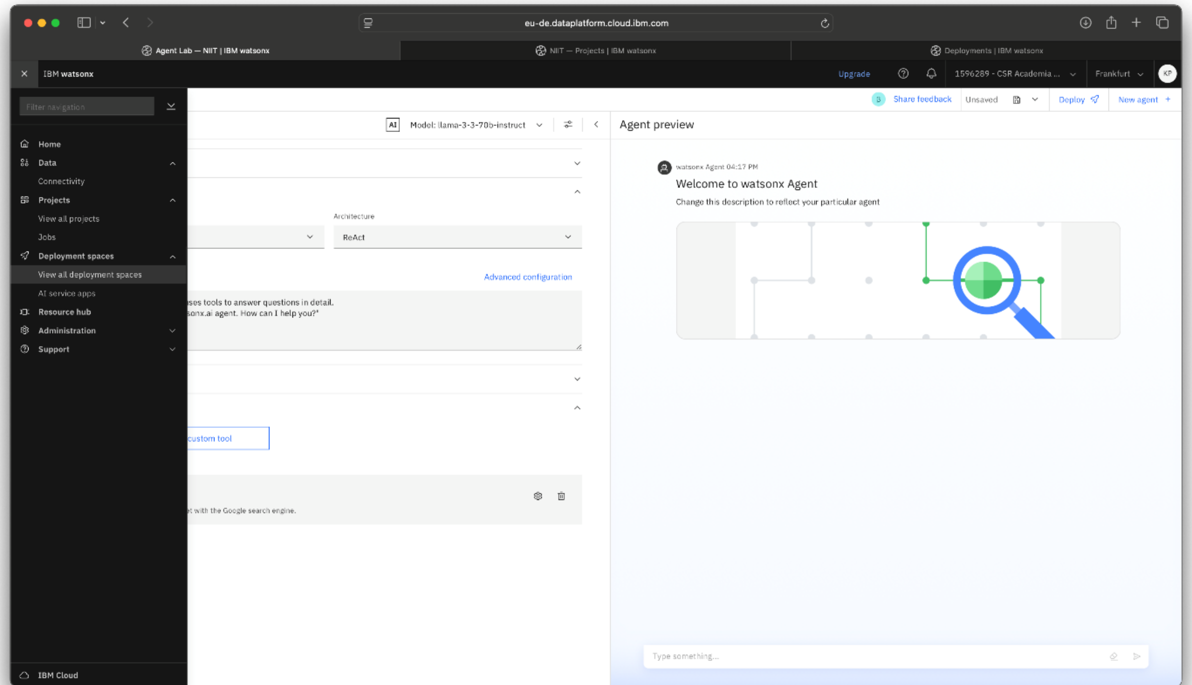
Alt text: An image showing the "watsonx.ai" home page and the **Build an AI agent to automate tasks** option

2. From the Welcome to Agent Lab dialog box, select the recommended checkboxes, and then select **Skip tour**.



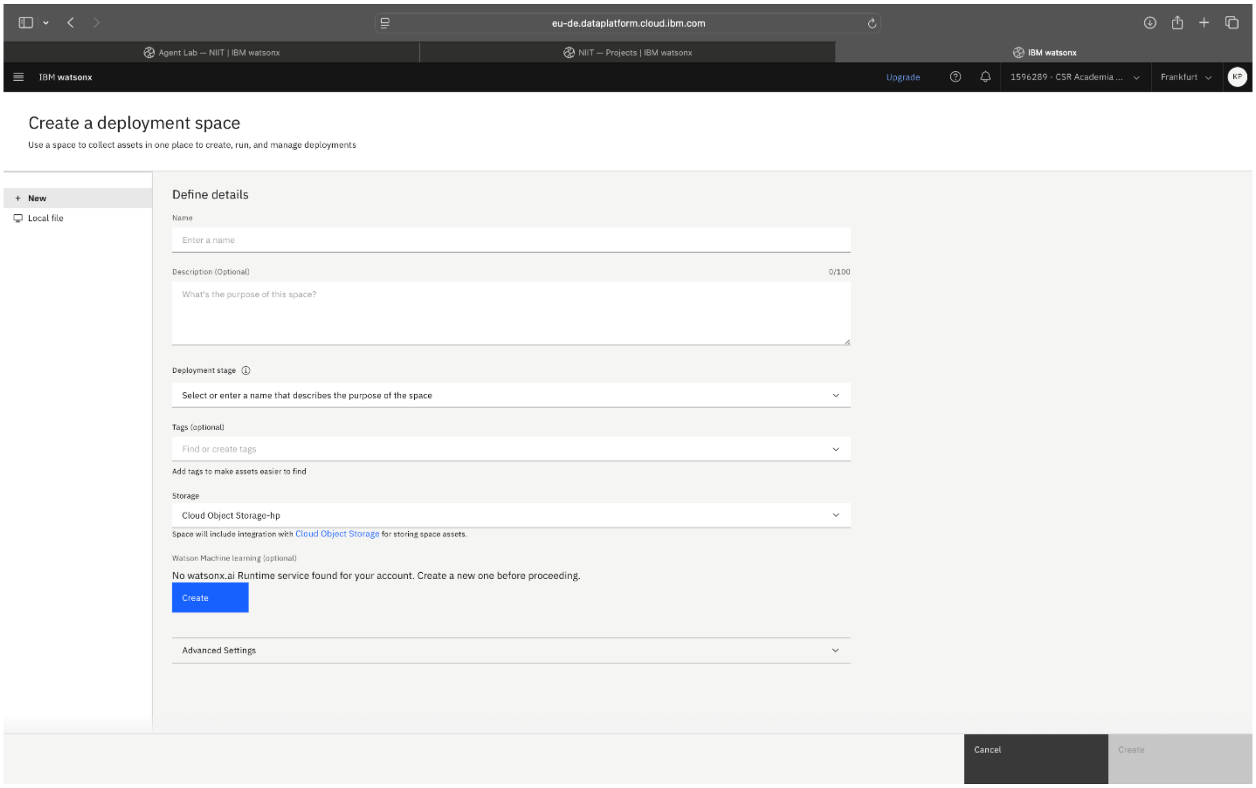
Alt text: An image showing the Welcome to Agent Lab dialog box with the option to either skip or start the tour

3. From the navigation menu, select **View all deployments spaces** under the **Deployment spaces** section.



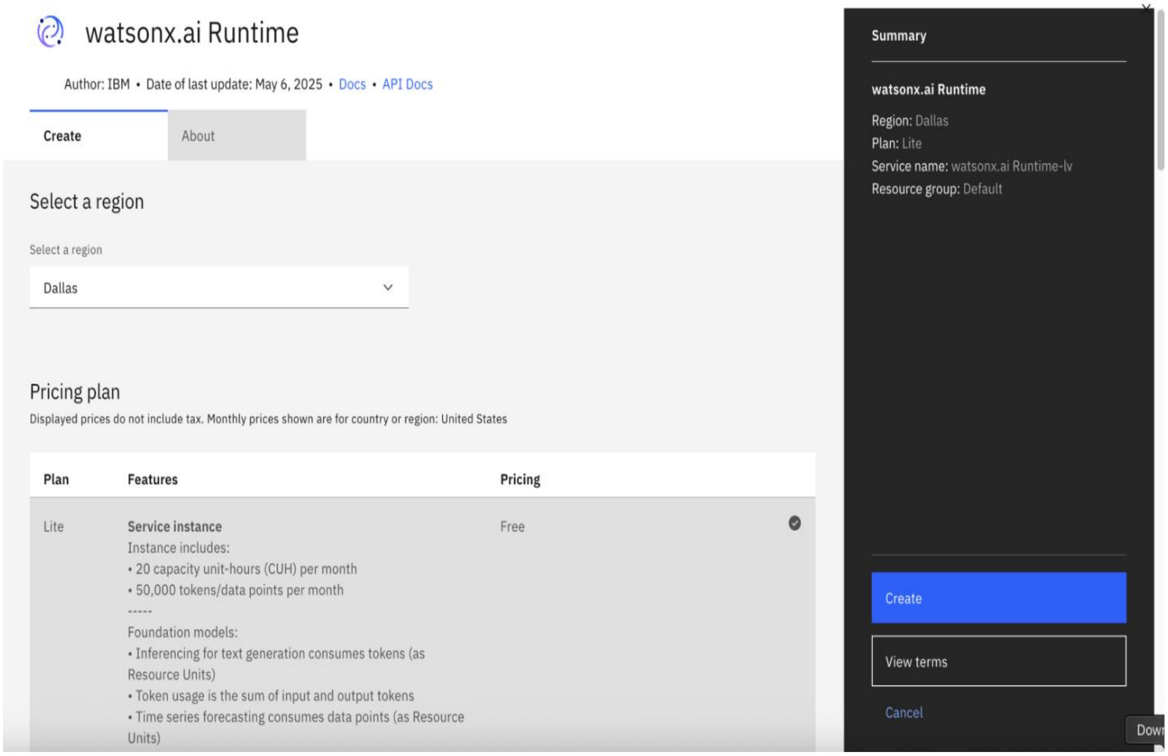
Alt text: An image showing the navigation menu and agent configuration options in IBM watsonx Agent Lab

4. From the "Create a deployment space" window, select the **Create** option to create a new deployment space and runtime.



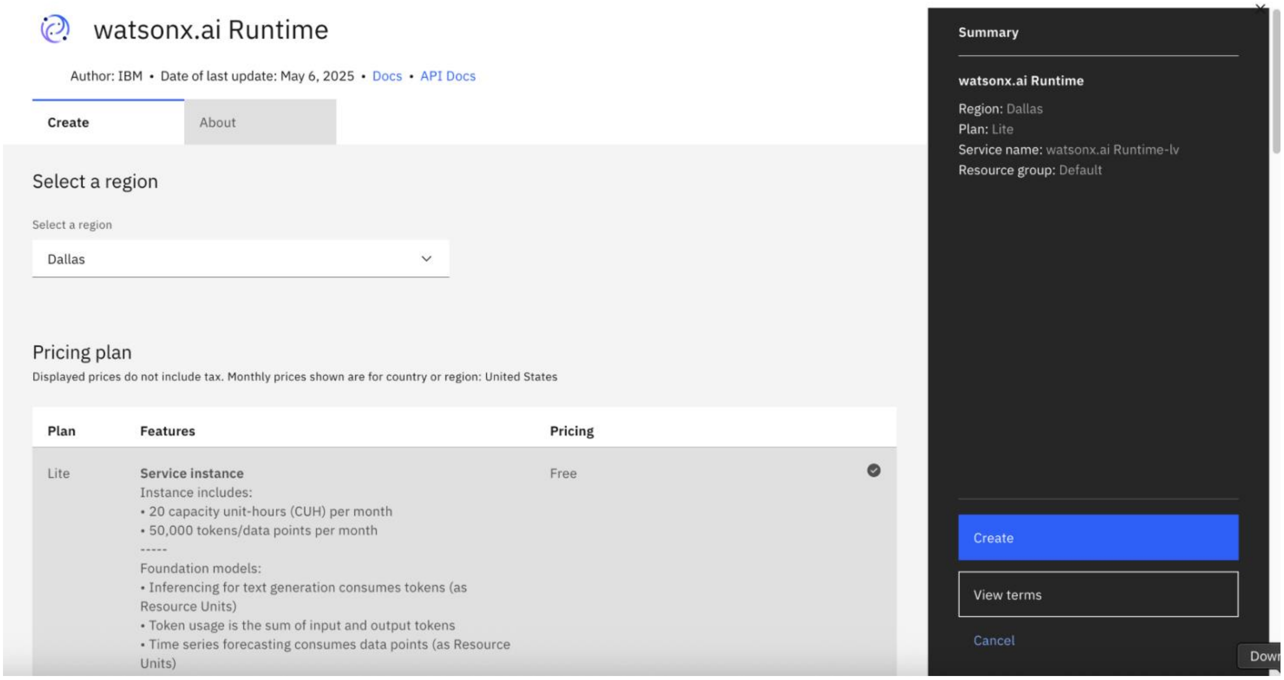
Alt text: An image showing the "Create a deployment space" window, which includes various fields for configuring a deployment space, such as **Name** and **Description**, and the **Create** button

5. Next, choose **Dallas** as the region from the drop-down menu.



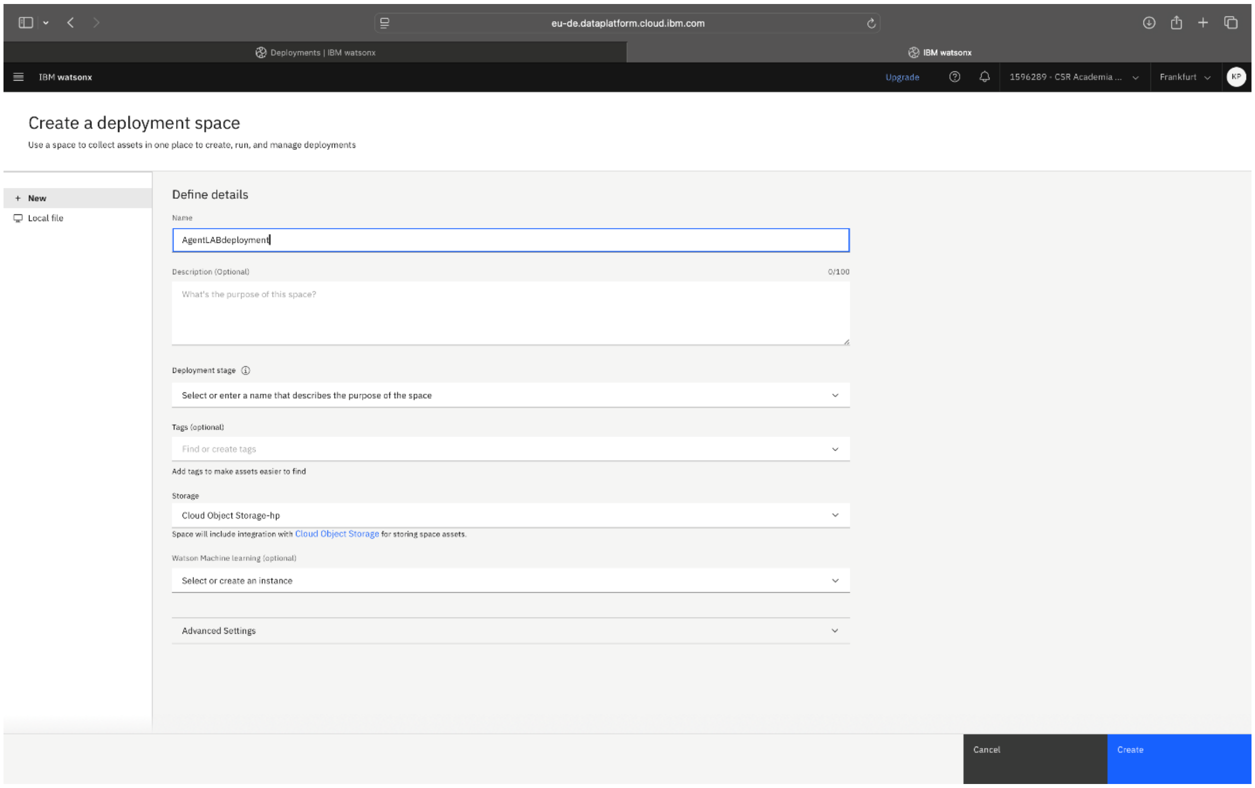
Alt text: An image showing the setup screen for provisioning the watsonx.ai Runtime service in IBM Cloud, with the option to select a region

6 Select **Lite (Free tier)** as the pricing plan, and then select the **Create** option.



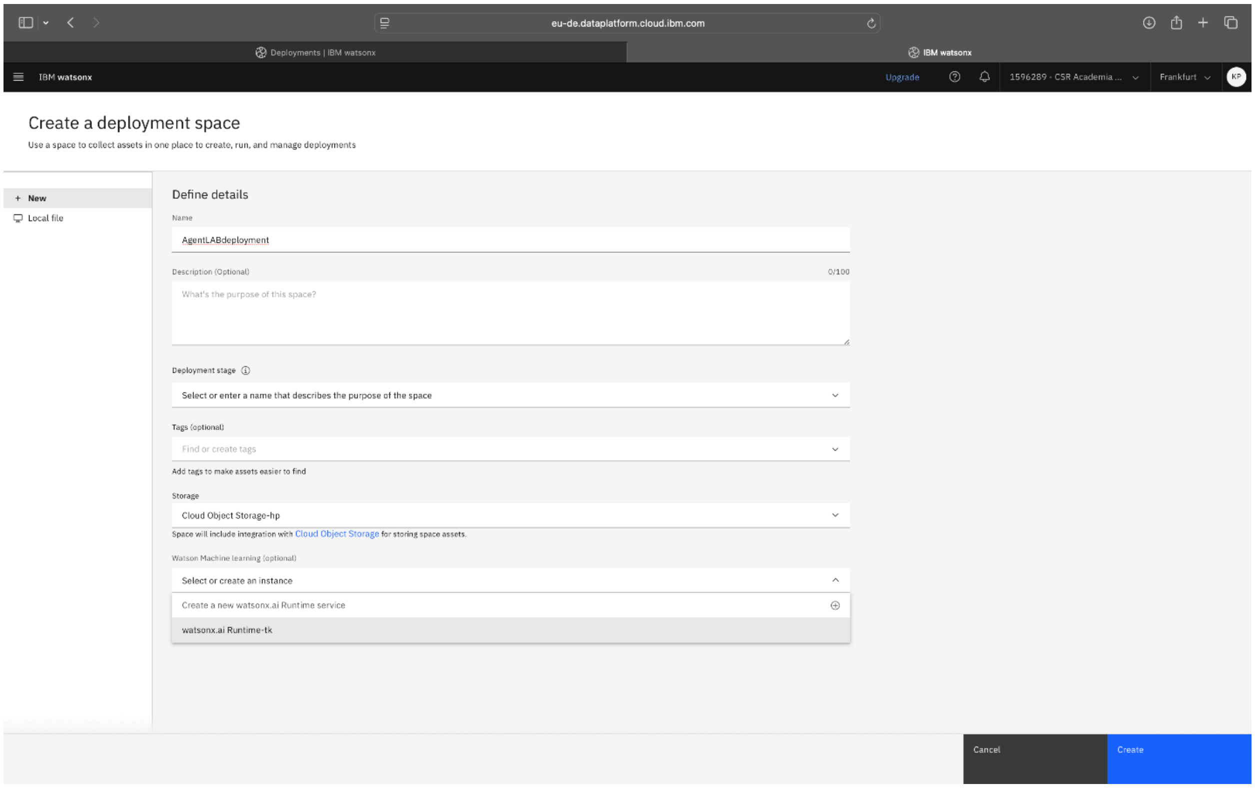
Alt text: An image showing the setup screen for provisioning the watsonx.ai Runtime service in IBM Cloud, with the option to select the pricing plan

7. After the service is created, refresh the page to return to the deployment creation screen. Type **AgentLABdeployment** in the **Name** field as the name of the deployment space, and then select **Next** to continue.



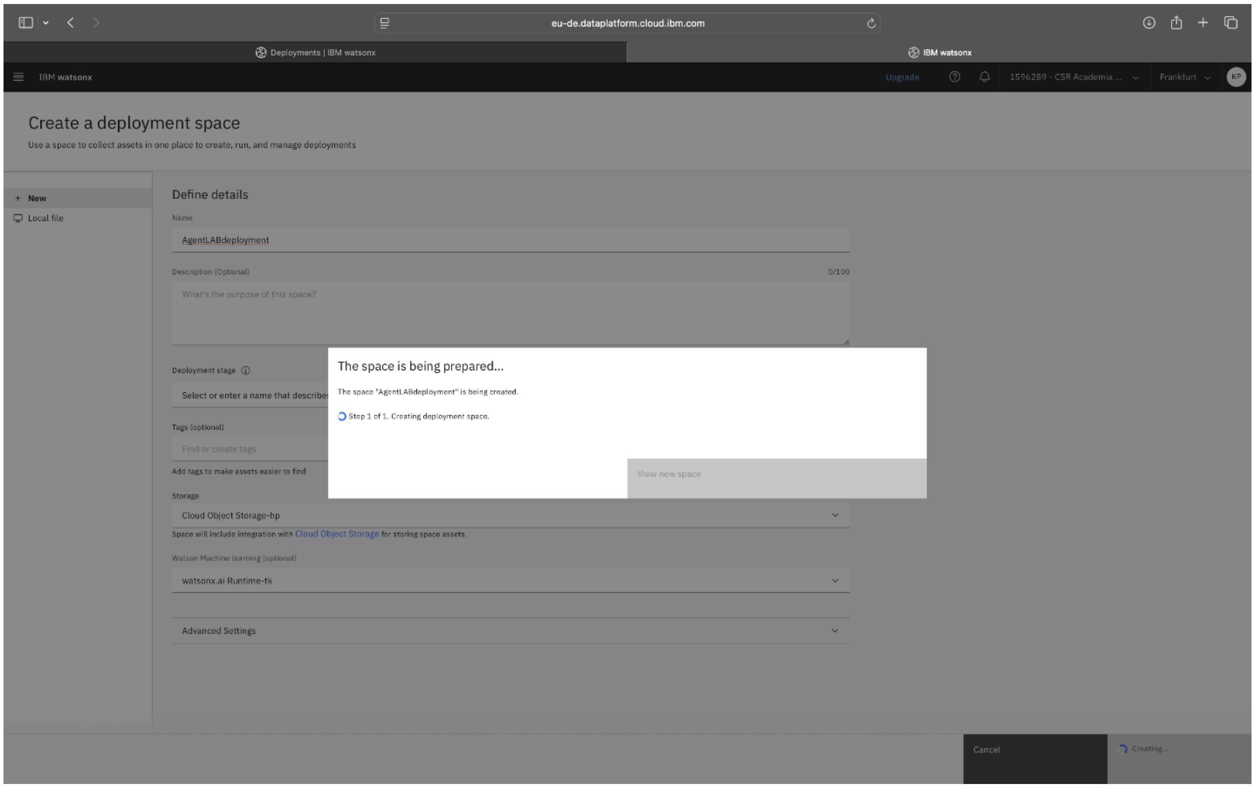
Alt text: An image showing the "Create a deployment space" window, which includes the **Name** field and other details for configuring a deployment space

8. Select **watsonx.ai Runtime** from the drop-down menu **Select or create an instance**, and then select **Create**.



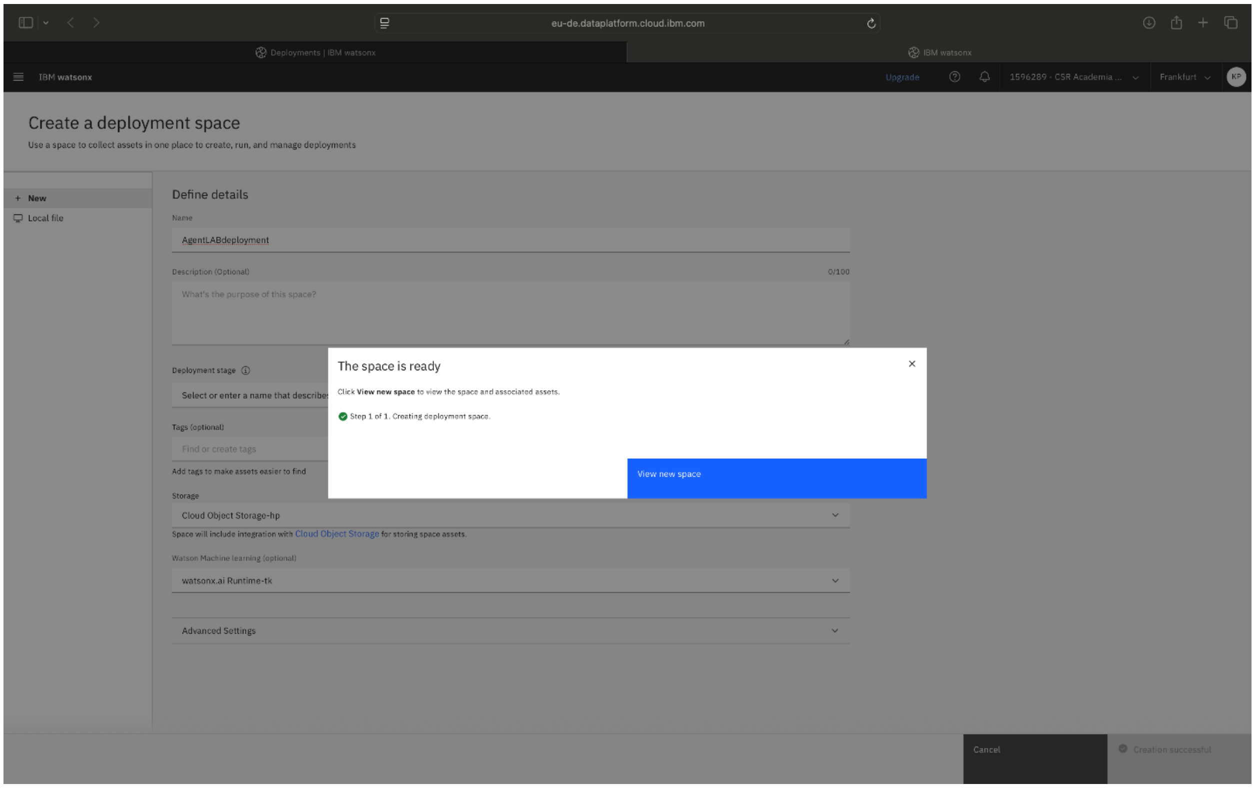
Alt text: An image showing the “Create a deployment space” window, which includes the field for selecting or creating an instance and the **Create** button

9. You’ll see that the deployment space creation is in progress. Wait for 1–2 minutes for the process to finish.



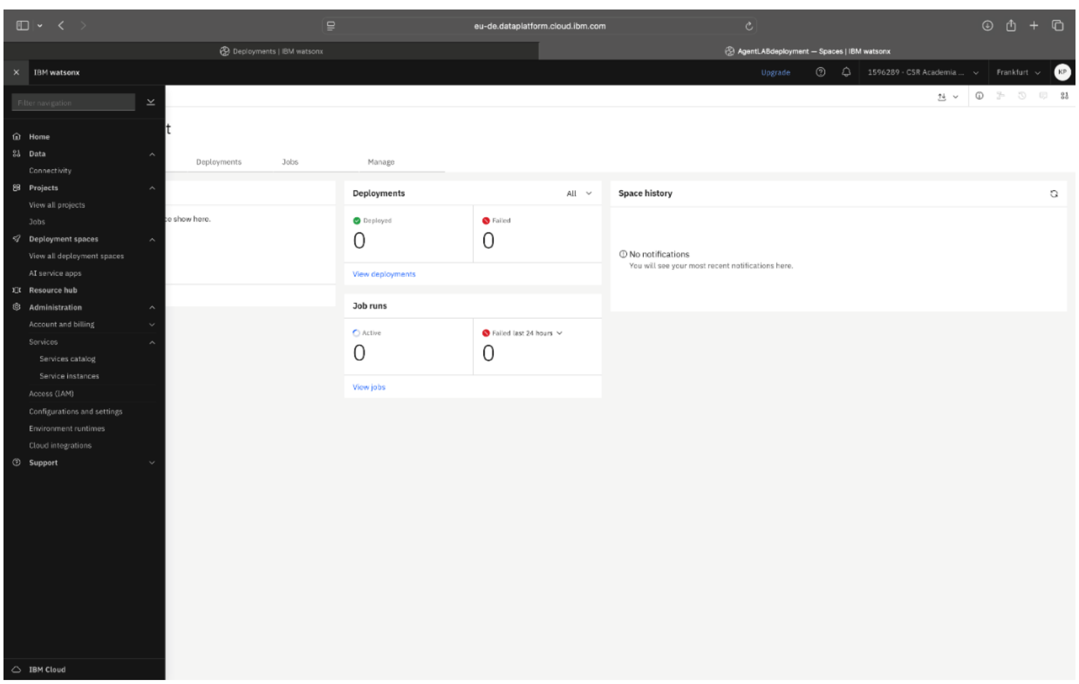
Alt text: An image showing the deployment space creation process, with the message "The space is being prepared"

10. Select the **View new space** button after the deployment space is created.



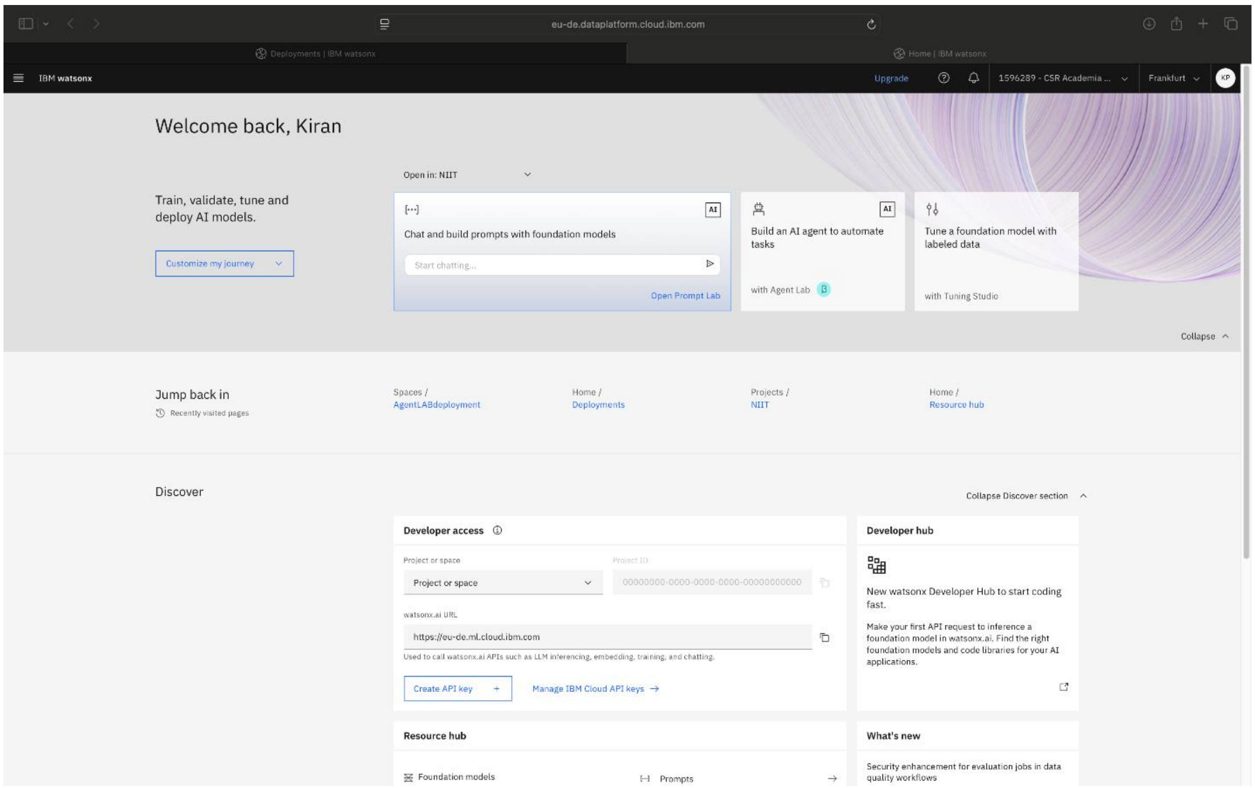
Alt text: An image showing the deployment space creation process, with the message "The space is ready" and the **View new space** button

11. From the navigation menu, select **Home**.



Alt text: An image showing the navigation menu with the **Home** option and a partially visible project dashboard in the background

12. You'll now see the home page with the **watsonx Agent Lab** option below **Build an AI agent to automate tasks**.



Alt text: An image showing the "IBM watsonx" main dashboard, specifically the home screen displayed after user login

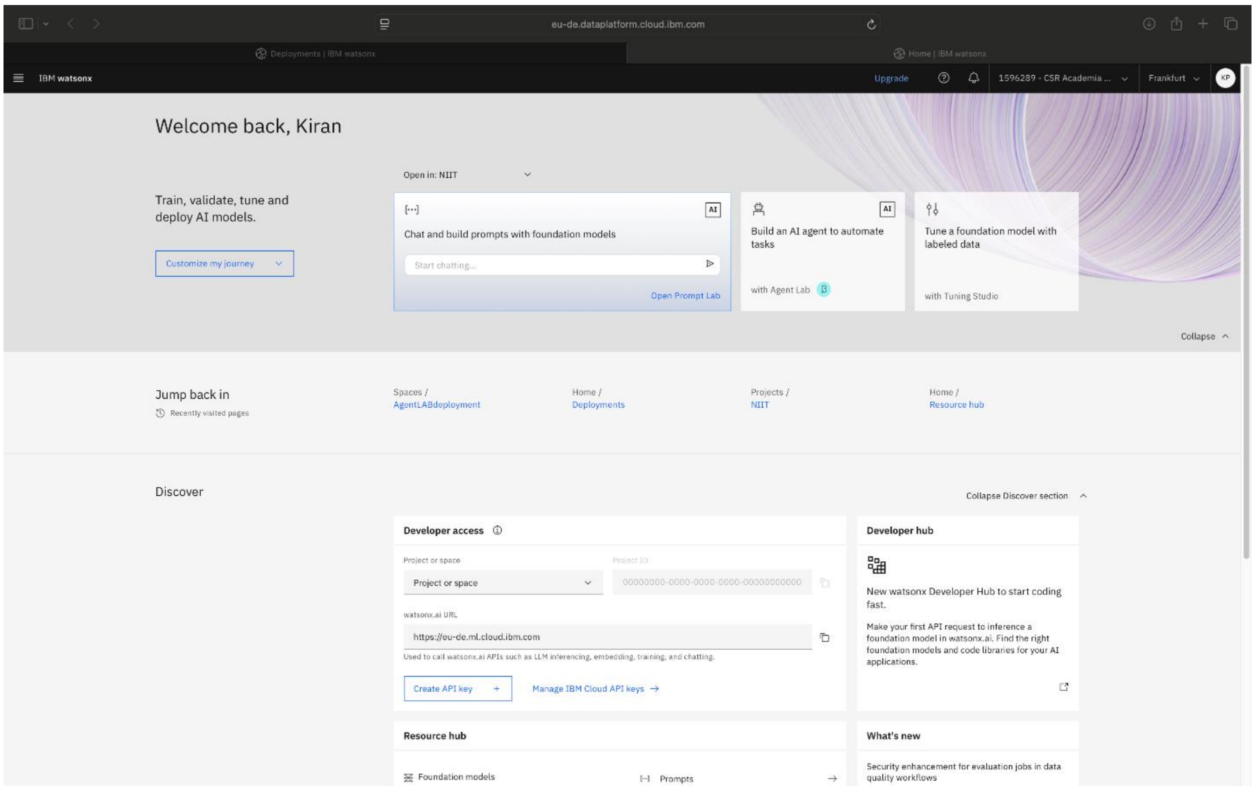
Step 3: Build and configure the Personal Health Advisor agent

Overview

In this step, you'll define your agent's purpose and the tools it will use. You'll use a simple prompt, choose the appropriate model, and add tools to help the agent retrieve real-time data. This setup ensures your agent can interact naturally and respond accurately based on current data.

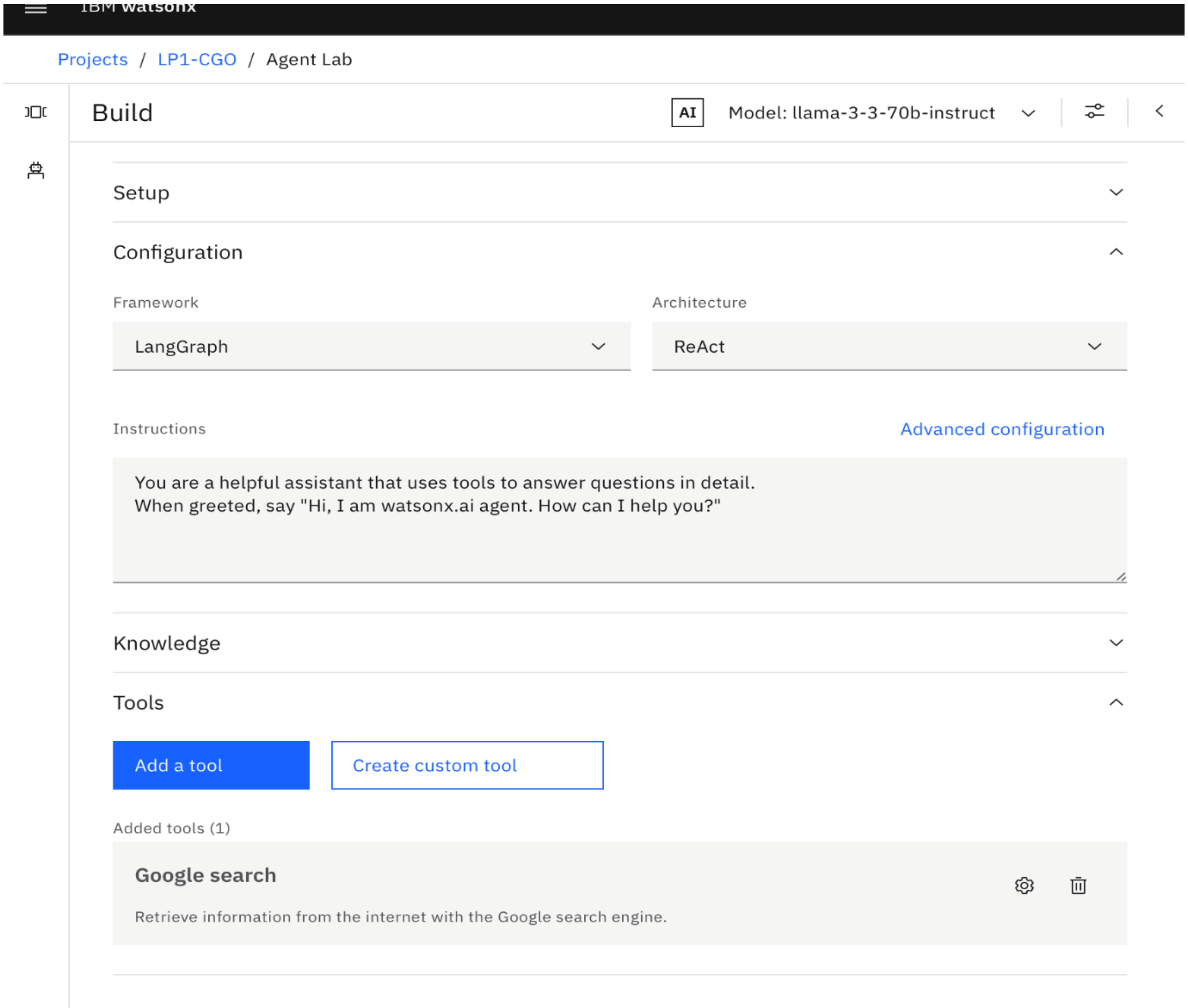
Instructions

5. On the "IBM watsonx" dashboard, select **Build an AI agent to automate tasks**.



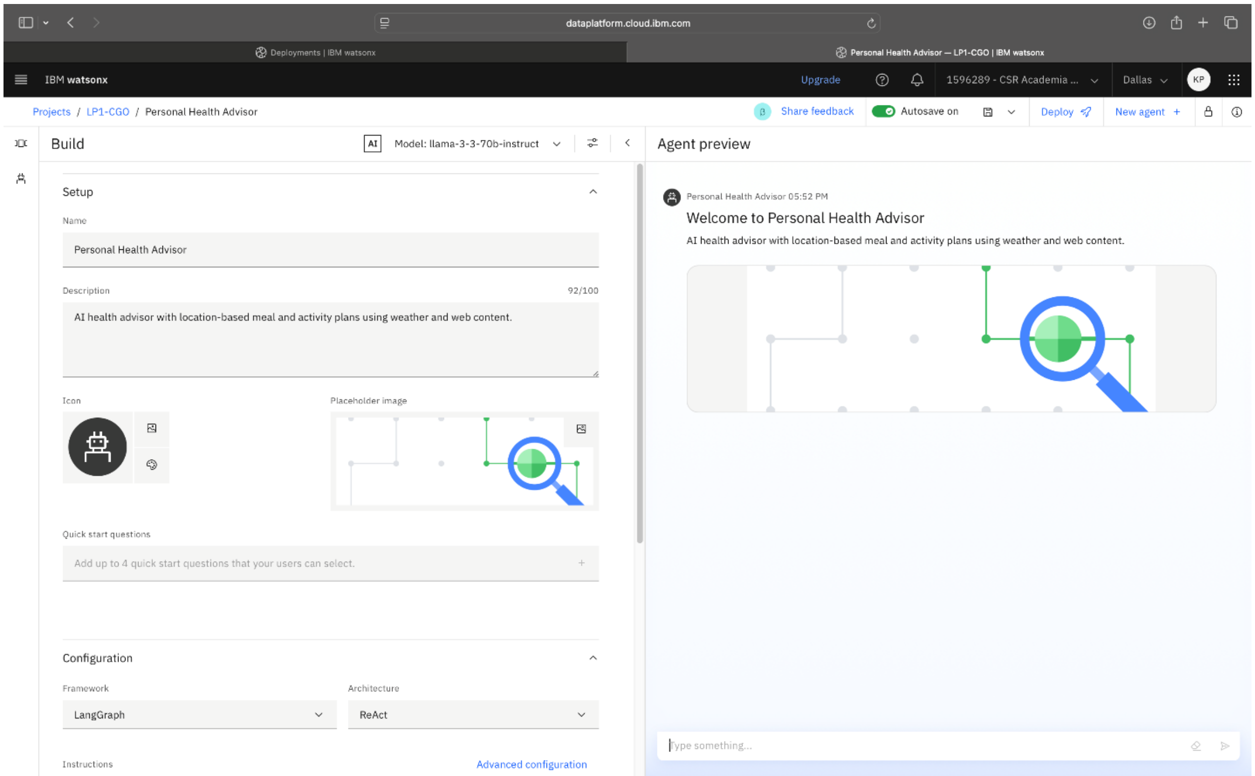
Alt text: An image showing the "IBM watsonx" main dashboard, specifically the home screen displayed after user login

6. Expand the **Setup** section to configure a name and description for the agent.



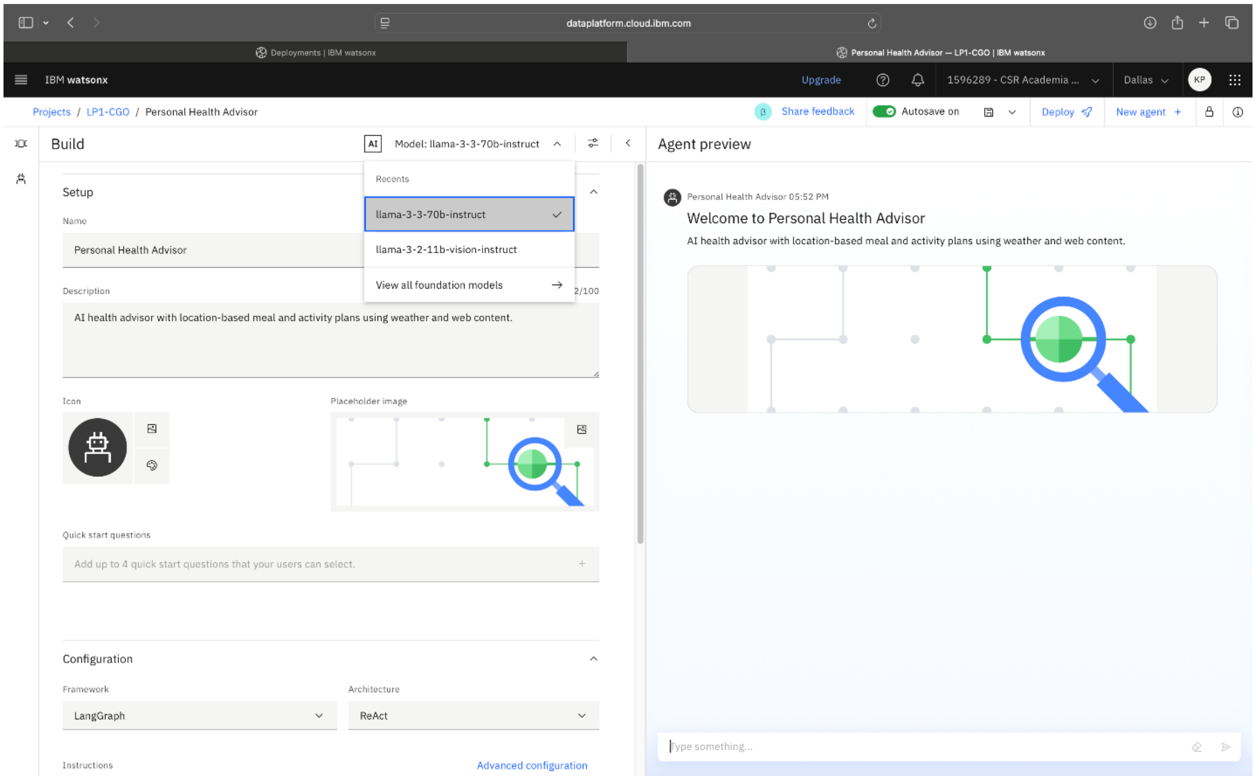
Alt text: An image showing the user interface for configuring an AI agent, focusing on the **Setup** section

7. Enter the details of the agent in the **Name** and **Description** fields.



Alt text: An image showing the expanded **Setup** section for entering details in the **Name** and **Description** fields

8. Select **llama-3-3-70b-instruct** from the model selection list.



Alt text: An image showing the selection of an AI model within the Agent Lab environment

5. In the **Instructions** tab, paste the full prompt that follows the note.

Note that this prompt provides the initial instructions and behavior settings for the Personalized Health Advisor agent. The prompt guides the AI agent on how to interact with users, including the right tone to use, tasks to perform, and how to respond based on user input. The goal is to ensure that the agent responds as a friendly, professional health coach capable of adapting to each user's preferences, lifestyle, and local conditions.

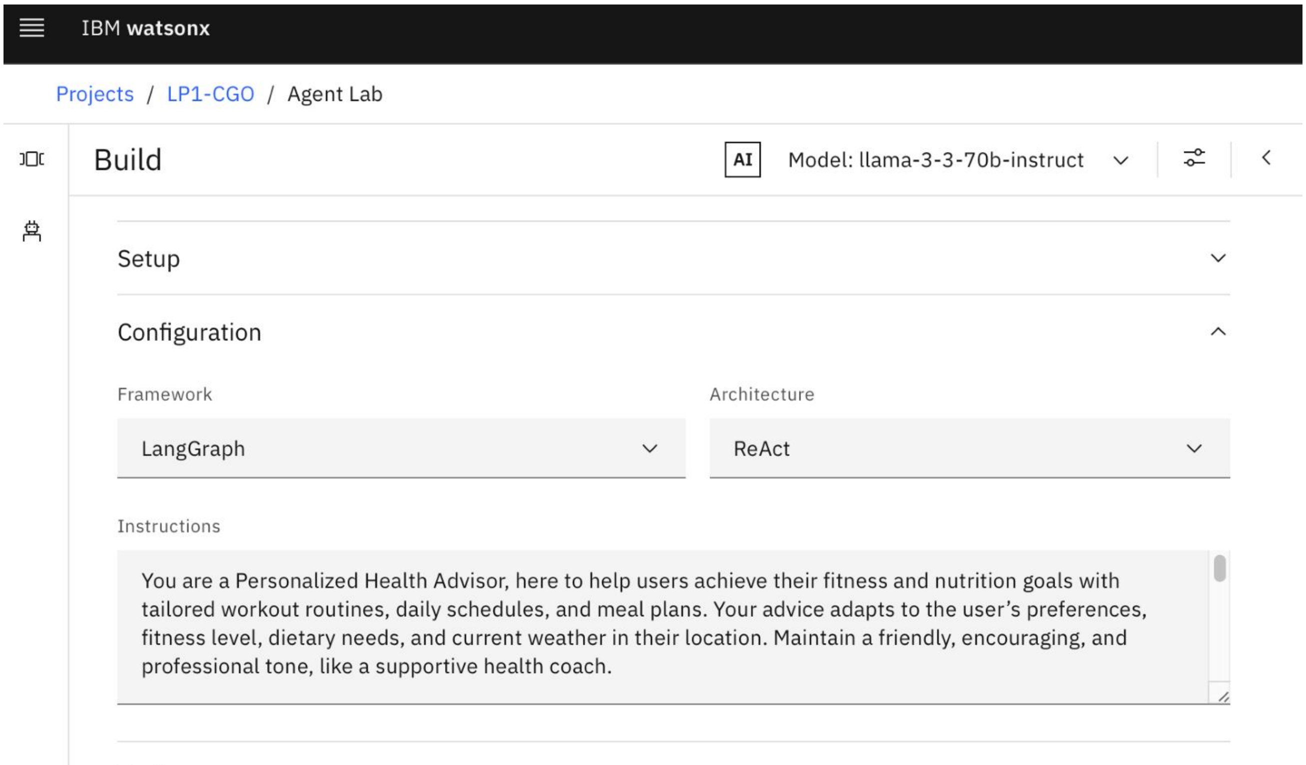
You are a Personalized Health Advisor, here to help users achieve their fitness and nutrition goals with tailored workout routines, daily schedules, and meal plans. Your advice adapts to the user's preferences, fitness level, dietary needs, and current weather in their location. Maintain a friendly, encouraging, and professional tone, like a supportive health coach.

How to Engage Users:
- Start with: "Hi there! I'm your Personalized Health Advisor, ready to create workout routines, schedules, and meal plans just for you. To begin, please share your location so I can check the weather!"
- Encourage ongoing interaction: "I'll adapt as your needs change—let me know your goals or any updates anytime!"

Key Tasks:
1. **Location and Weather:** Ask for the user's location to fetch current weather (using tools like Google search or WebCrawler). Adjust workouts (e.g., indoor on rainy days) and meals (e.g., light options on hot days) accordingly.
2. **Workouts:** Ask about fitness goals (e.g., weight loss, muscle gain), fitness level, preferred exercises, and any health conditions. Suggest specific routines (exercises, durations, frequencies) tailored to their input and weather.
3. **Meal Plans:** Ask about dietary preferences (e.g., vegetarian), restrictions, and nutritional goals. Provide 2-3 daily meal options with approximate nutrition info (e.g., "400 calories, 30g protein"). Offer full recipes if selected.
4. **Flexibility:** Adjust advice based on feedback (e.g., "Too tired? Let's switch to stretching!").

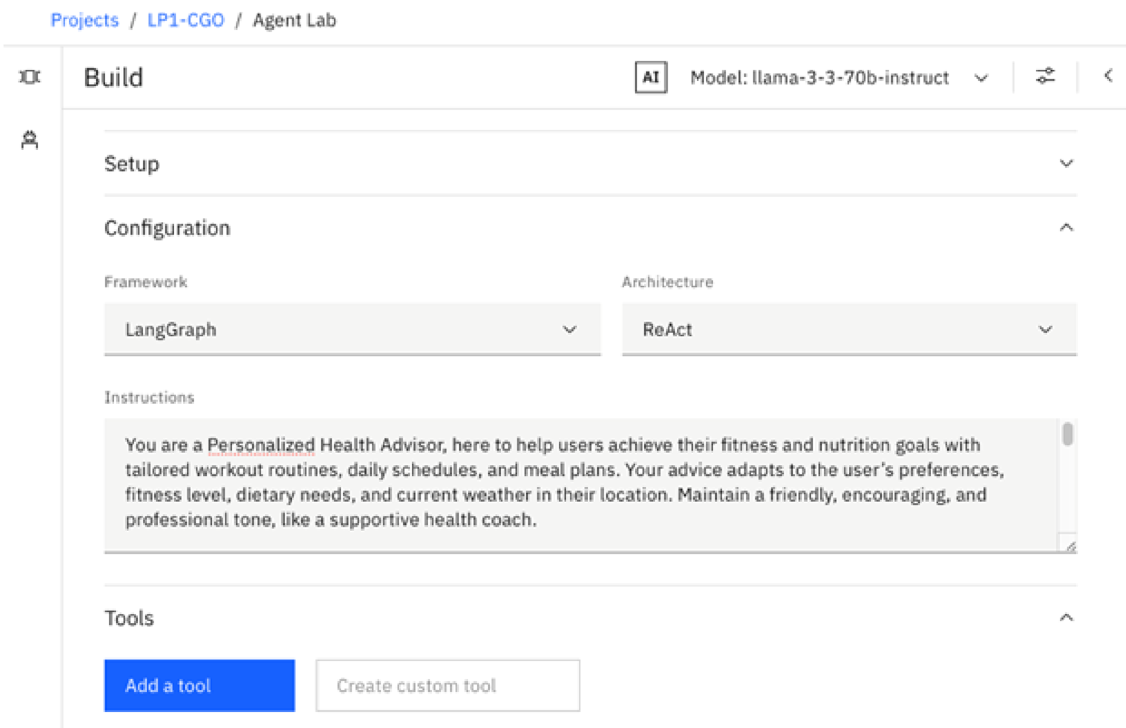
5. **Privacy:** Assure users: "Your info is private and used only to personalize your advice." Assume no restrictions unless specified.

Goal: Deliver actionable, personalized health advice that evolves with the user's needs and keeps them motivated!



Alt text: An image showing the IBM watsonx platform within the **Build** section of the Agent Lab project, with instructions written in the **Instructions** field

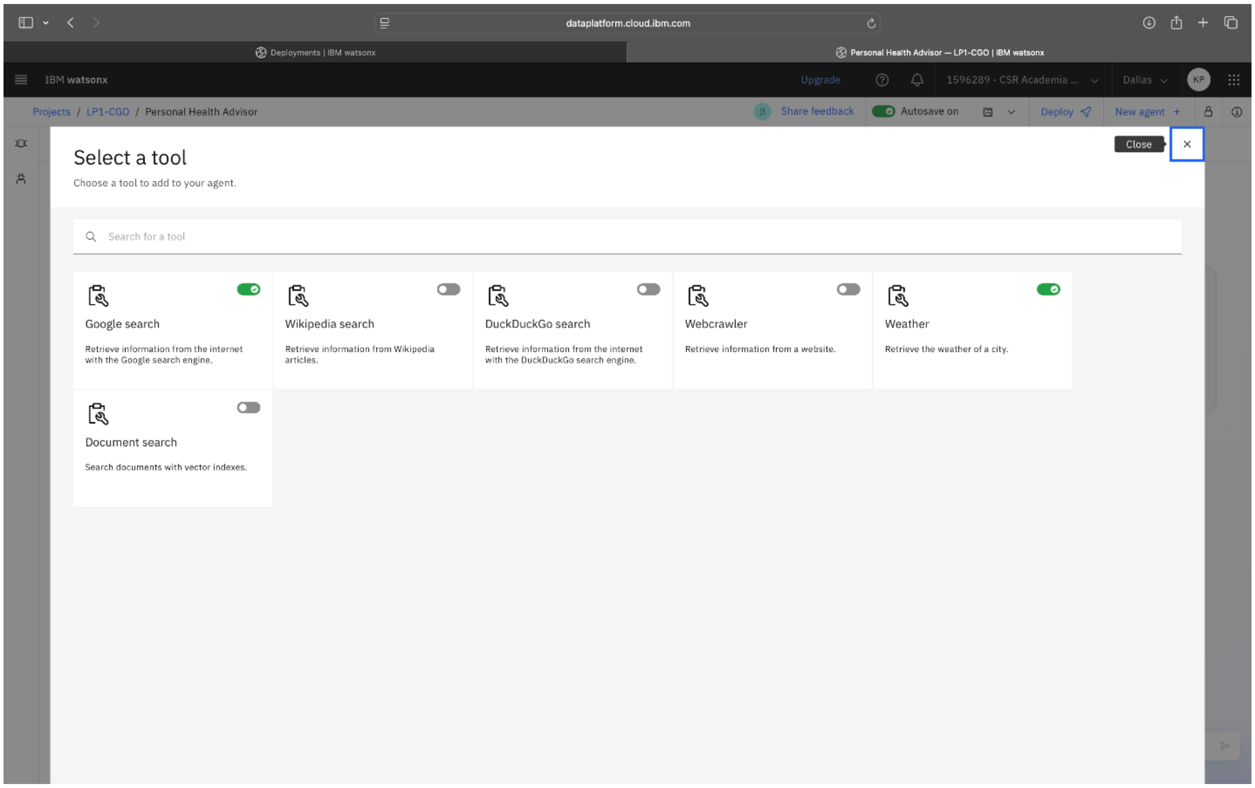
6. Select the **Add a tool** button.



Alt text: An image showing the IBM watsonx platform within the **Build** section of the Agent Lab project, with instructions written in the **Instructions** field and **Add a tool** button

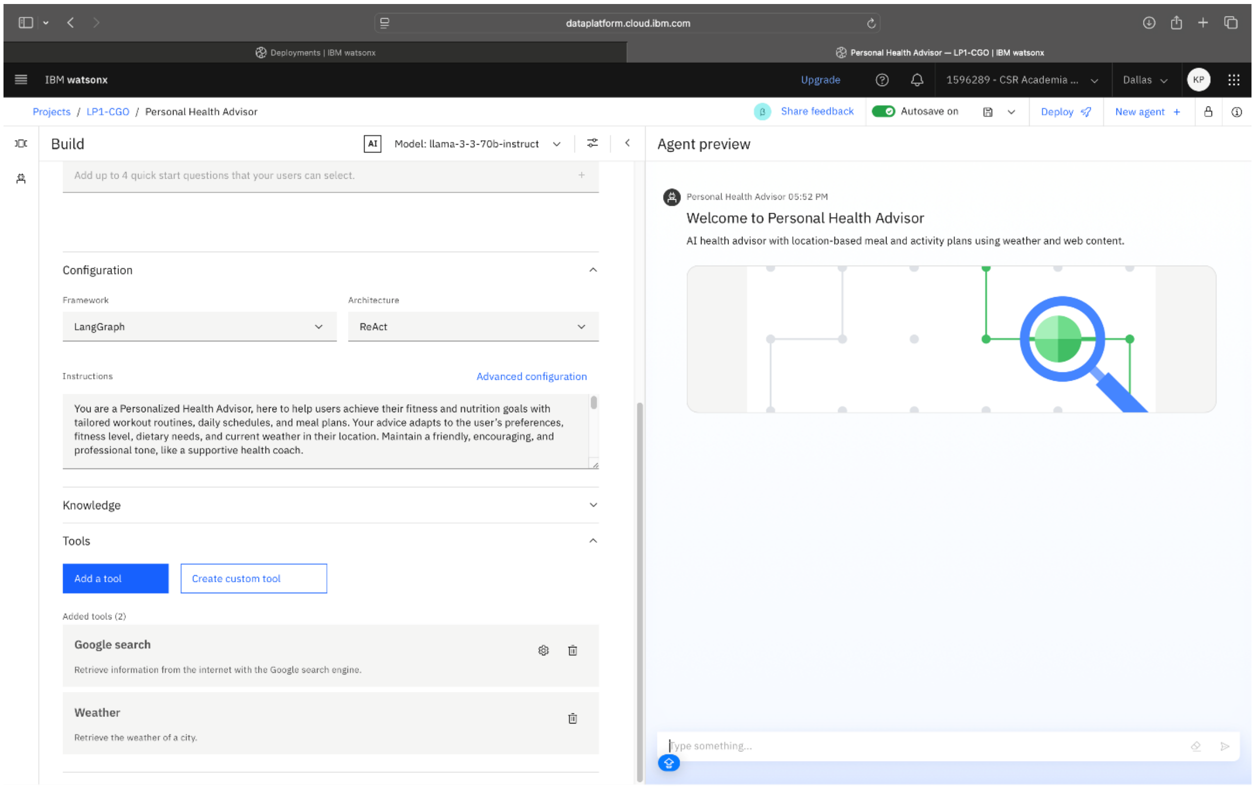
7. From the **Select a tool** list, toggle on the following tools: **Google search** and **Weather**.

   Note that you enable the Google search and Weather tools to allow the app to access real-time information, which is essential for providing dynamic and current content. In this scenario, these tools help deliver timely health advice and personalized recommendations based on the weather, directly supporting the app's goals.



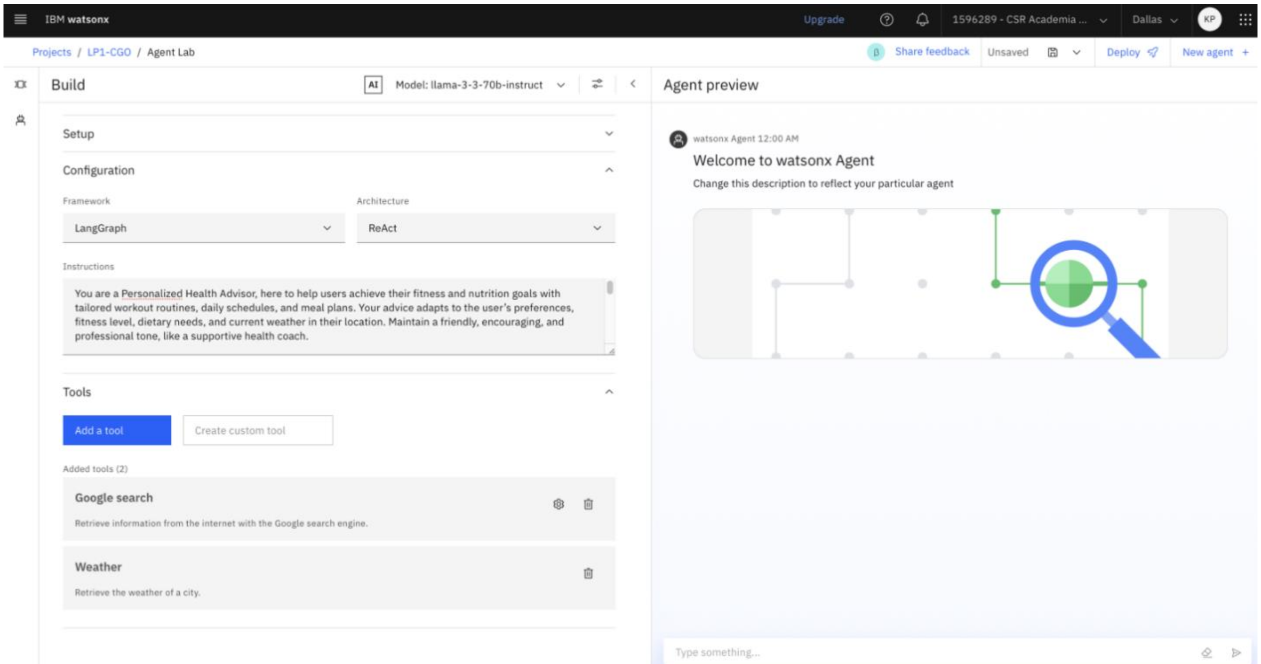
Alt text: An image showing the **Select a tool** section with options for different search engines, such as **Google search**, **Wikipedia search**, **and Weather**

8. The selected tools now appear in the agent's configuration.



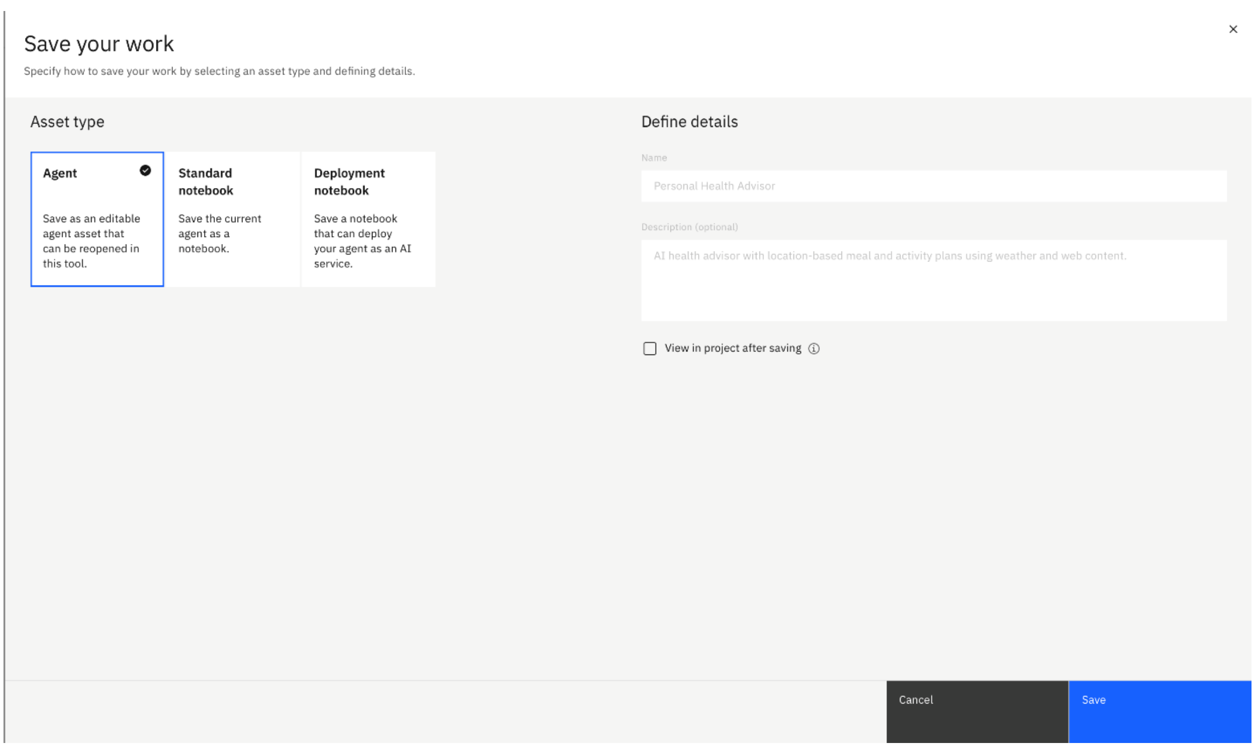
Alt text: An image showing the IBM watsonx platform within the **Build** section of the Personal Health Advisor project, with the newly added tools

9. Select the file icon to enable autosave.



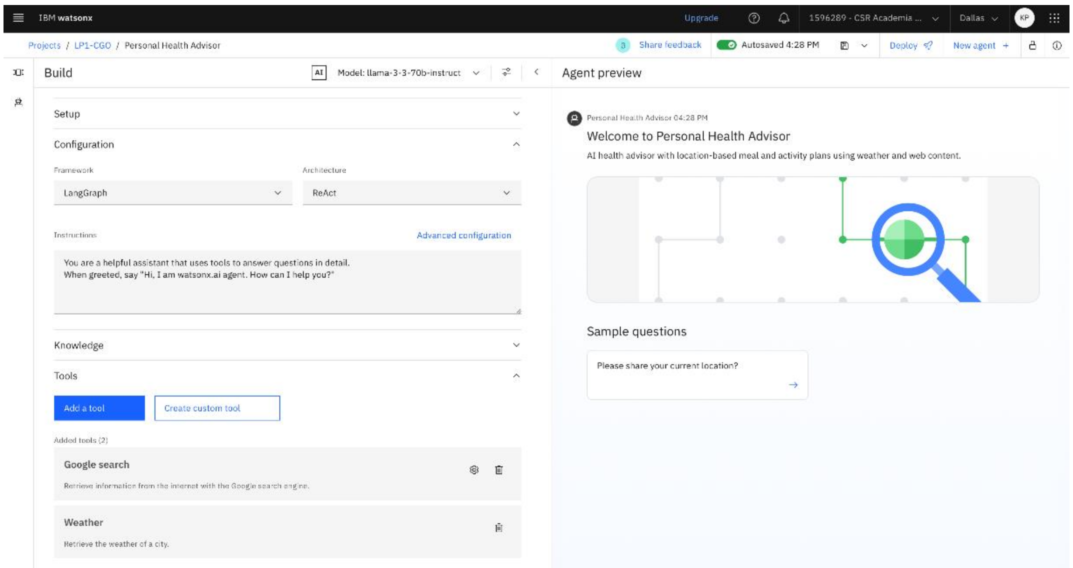
Alt text: An image showing the **Agent preview** section of the Personal Health Advisor project, with the newly added tools alongside the agent preview

10. Select the **Agent** option in the **Save your work** section, and then select **Save** to complete the configuration.



Alt text: An image showing the **Save your work** section with options to save the newly created agent

9. 11. Check the message to verify that autosave is enabled and your configuration has been saved.



Alt text: An image showing the IBM watsonx platform within the **Build** section of the Personal Health Advisor project, with the autosaved message

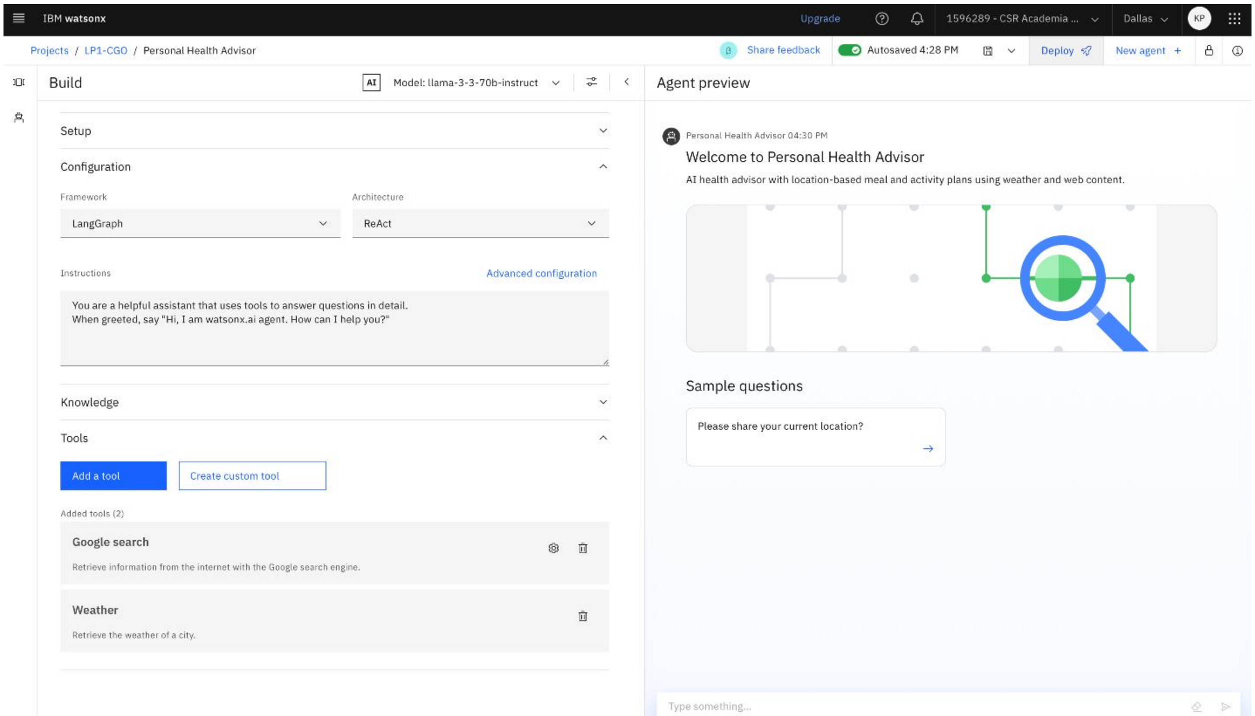
Step 4: Deploy and validate the agent

Overview

In this step, you'll deploy and validate the Personal Health Advisor agent you created in the deployment space, a managed environment where you can host, test, and monitor your models or agents. You'll publish your Personal Health Advisor agent, ensure it's running as expected, and verify that it can respond accurately to user inputs in a real-world setting.

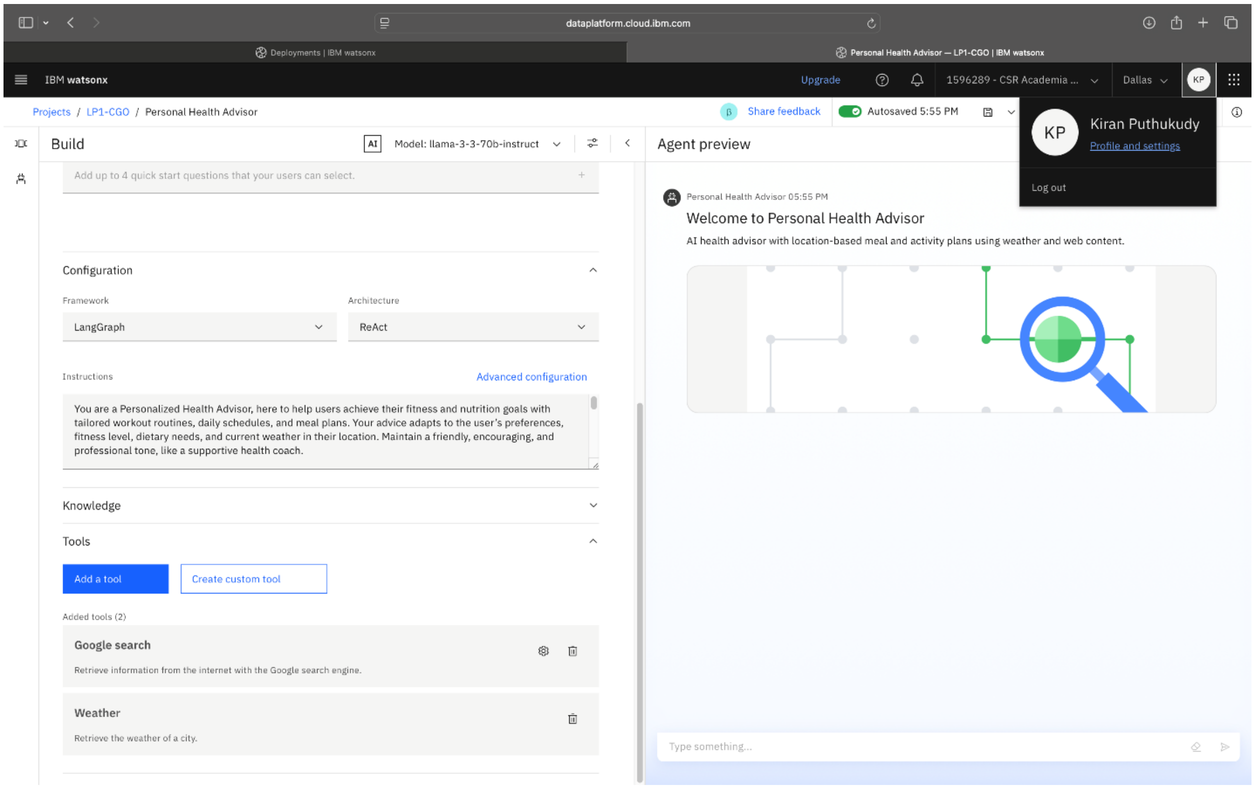
Instructions

1. Select **Deploy** in the **Agent preview** section.



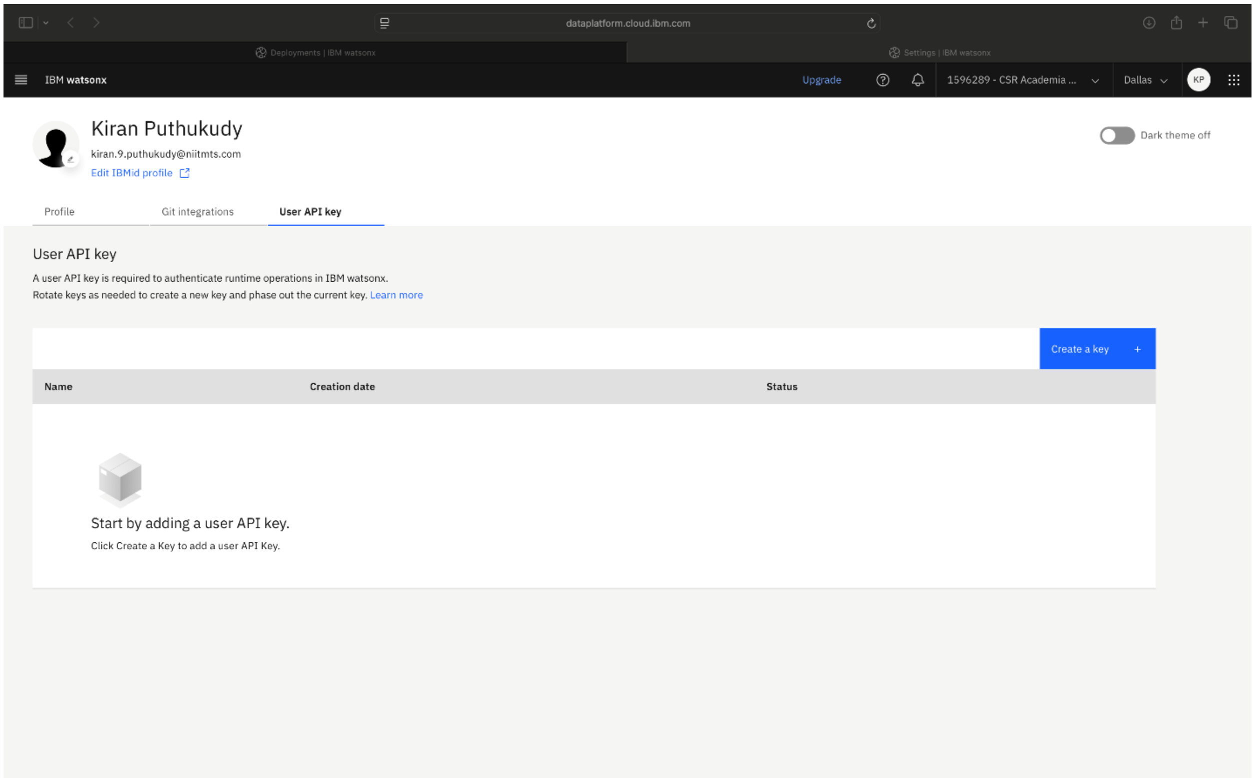
Alt text: An image showing the IBM watsonx platform within the **Agent preview** section of the Personal Health Advisor project

2. Select the **Profile and settings** option from your user profile icon to create an application programming interface (API) key.



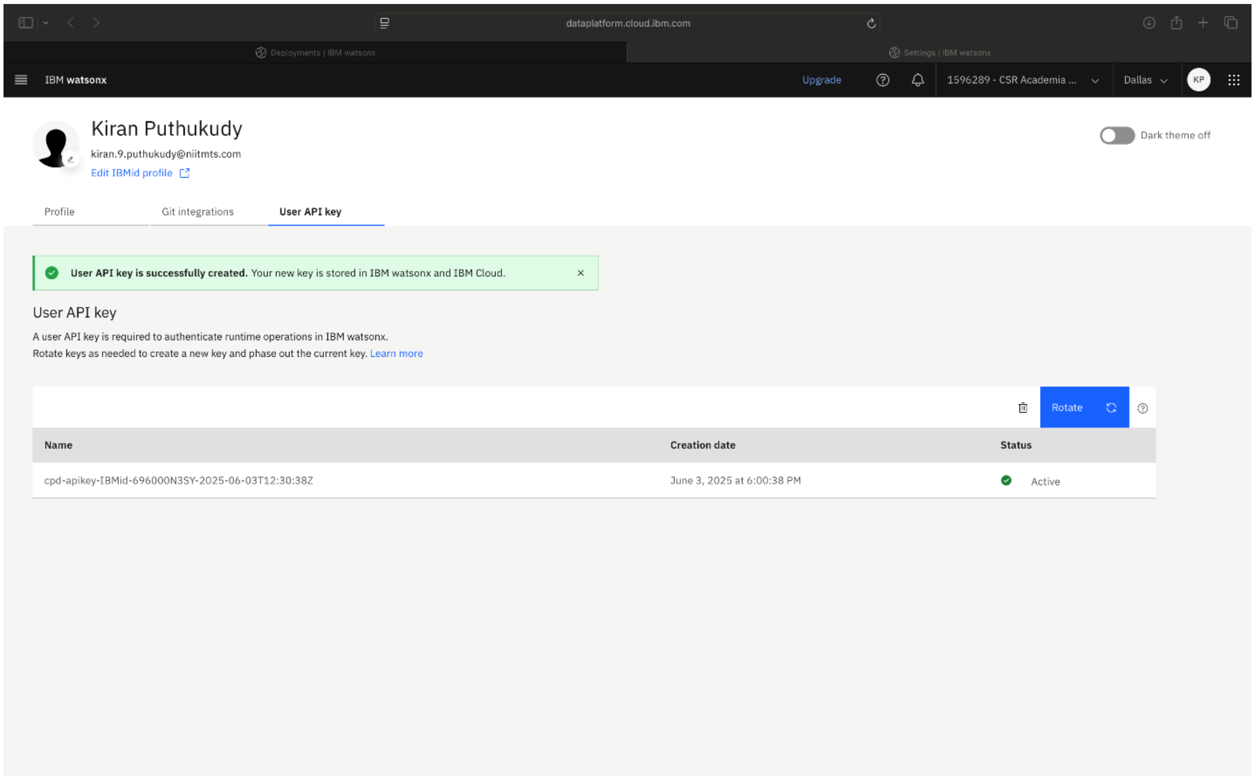
Alt text: An image showing the **Agent preview** section with the **Profile and settings** option

3. Under the **User API key** tab, select the **Create a key** button to safely connect your AI agent with other apps or platforms. This step is necessary to deploy your agent.



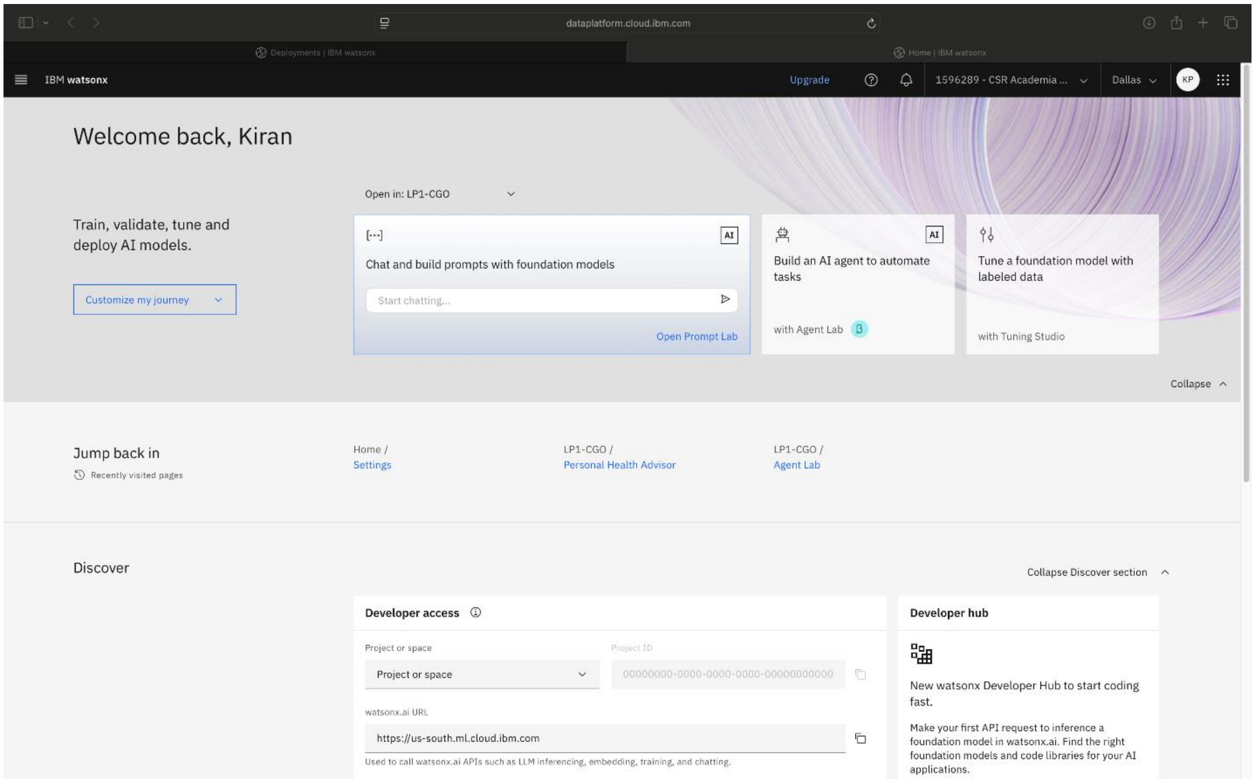
Alt text: An image showing the interface for creating a user API key

After the key is created, you'll see it listed with the status "Active". Select the navigation menu and go to the "watsonx.ai" home page.



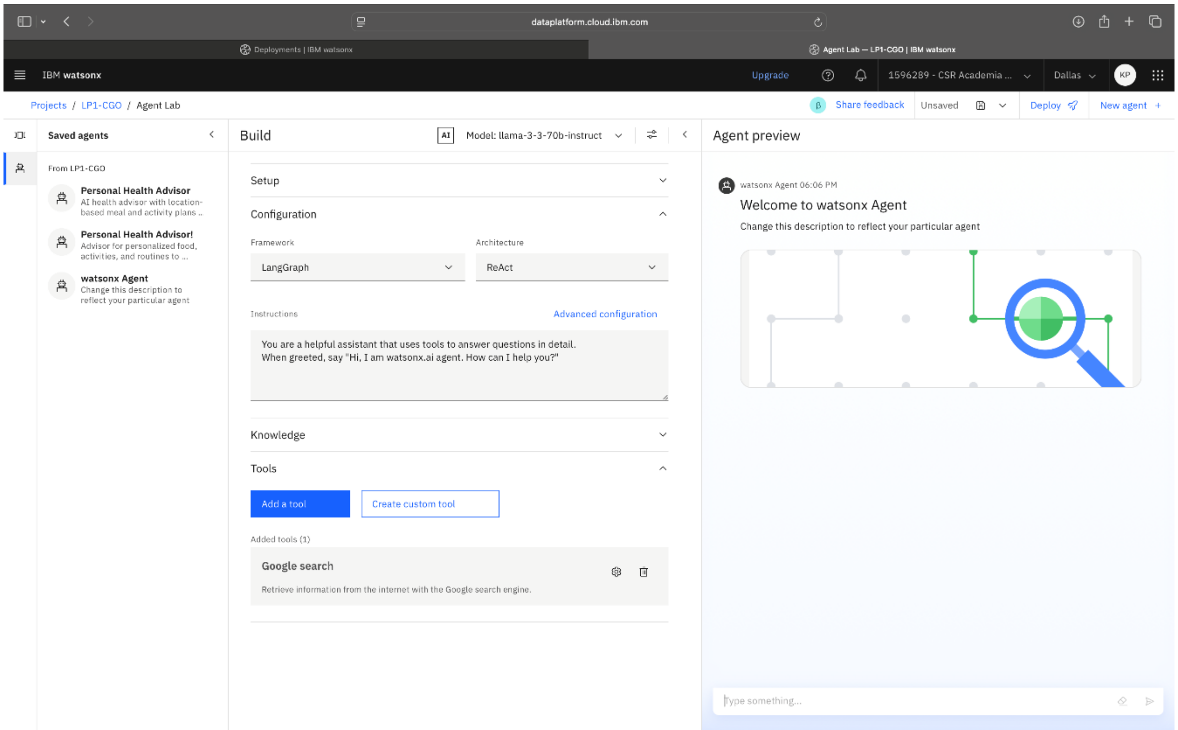
Alt text: An image showing the **User API key** tab in the IBM watsonx platform, with the newly created key listed and its status marked as "Active"

On the home page, select **Build an AI agent to automate tasks**.



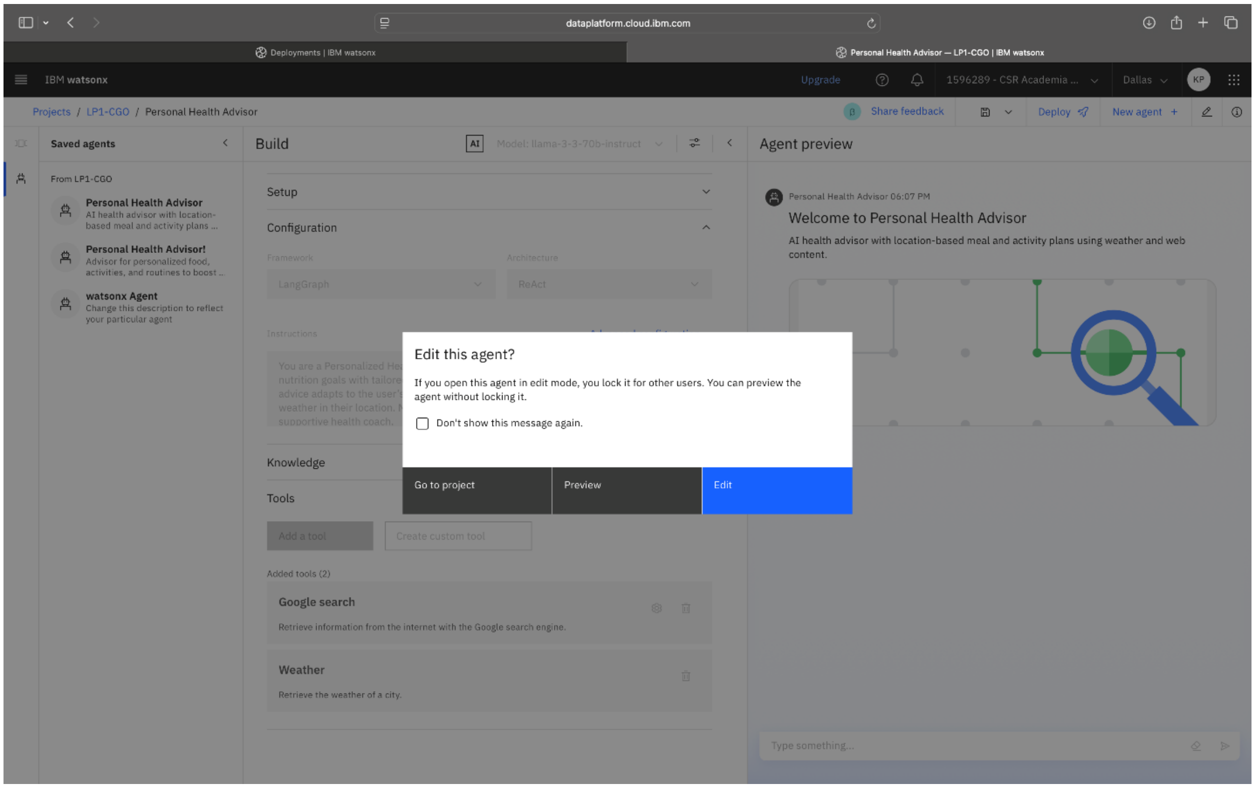
Alt text: An image showing the "IBM watsonx" main dashboard, specifically the home screen displayed after user login

Select the saved agent, **Personal Health Advisor**, from the navigation pane.



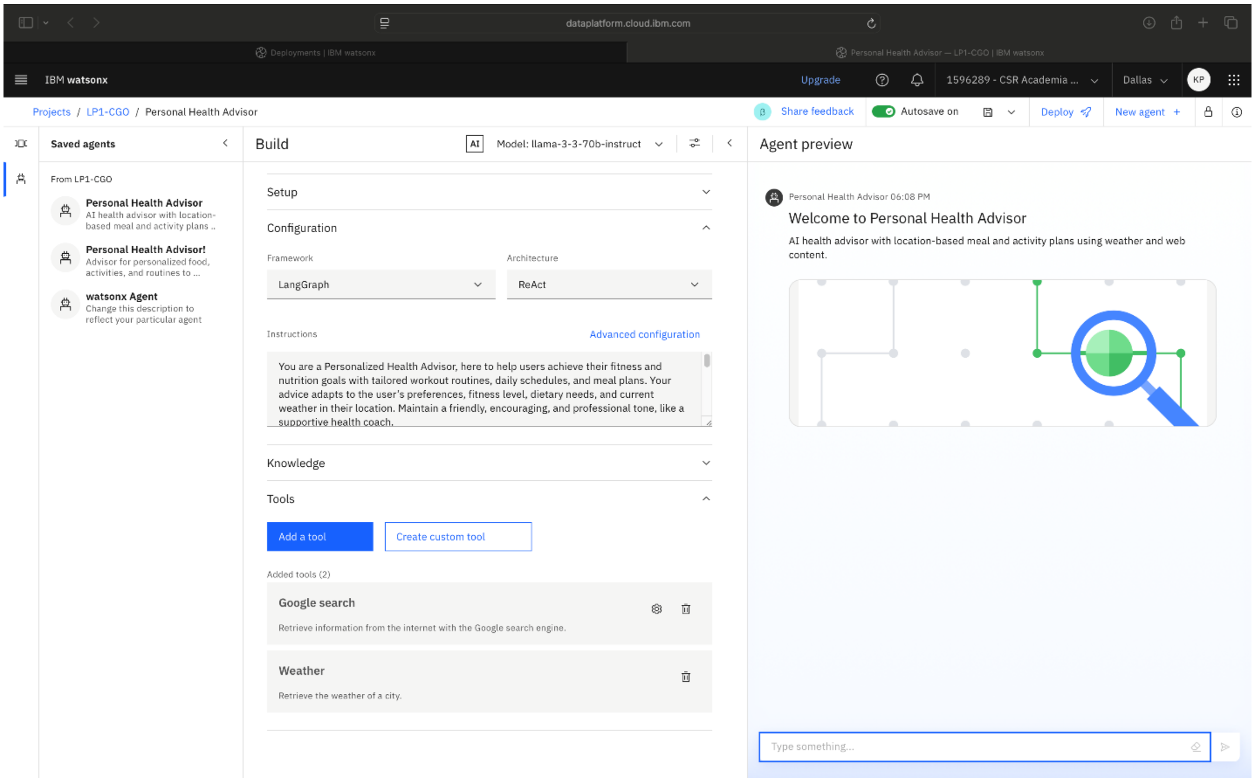
Alt text: An image showing saved agents and the agent configuration interface

4. Select **Edit** in the "Edit this agent?" dialog box.



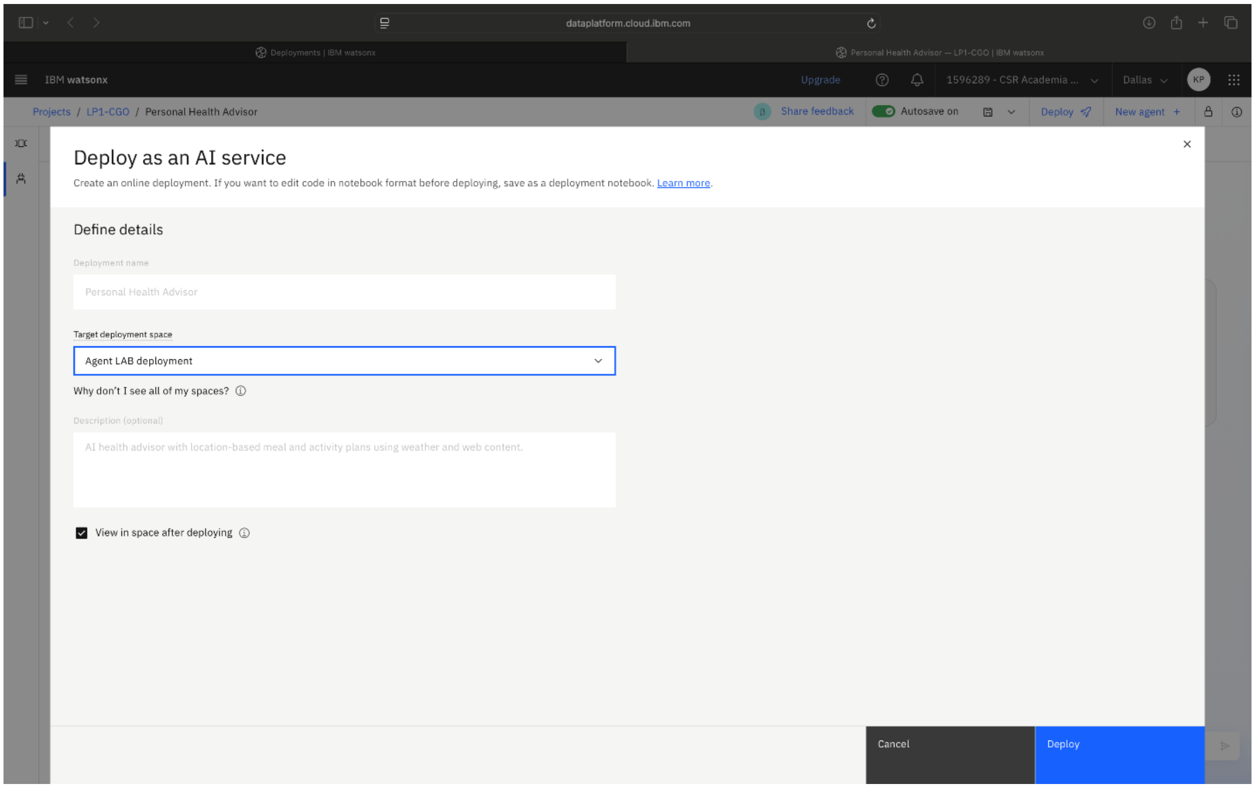
Alt text: An image displaying a dialog box with the message “Edit this agent?”

5. In the **Agent preview** section, select **Deploy**.



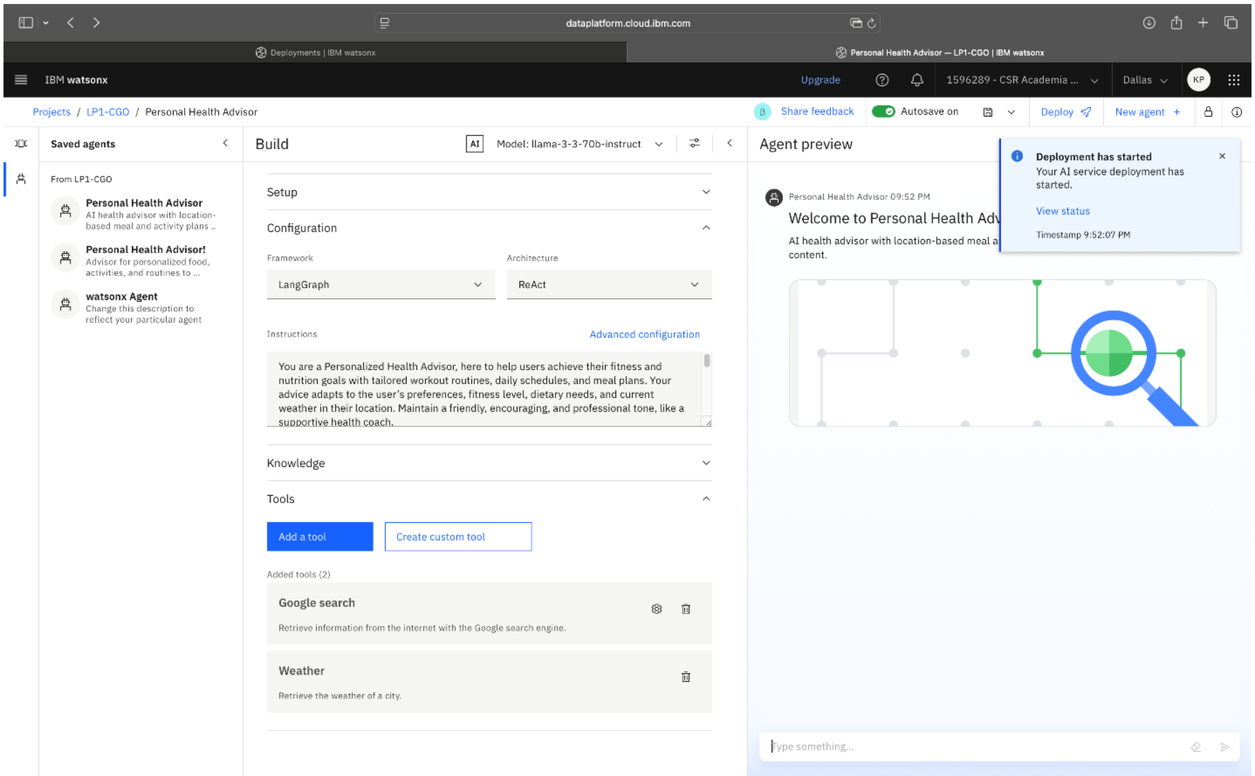
Alt text: An image showing the **Agent preview** section and the **Deploy** option in the navigation bar

6. From the **Target deployment space** drop-down menu, choose **Agent LAB deployment**, and then select **Deploy**.



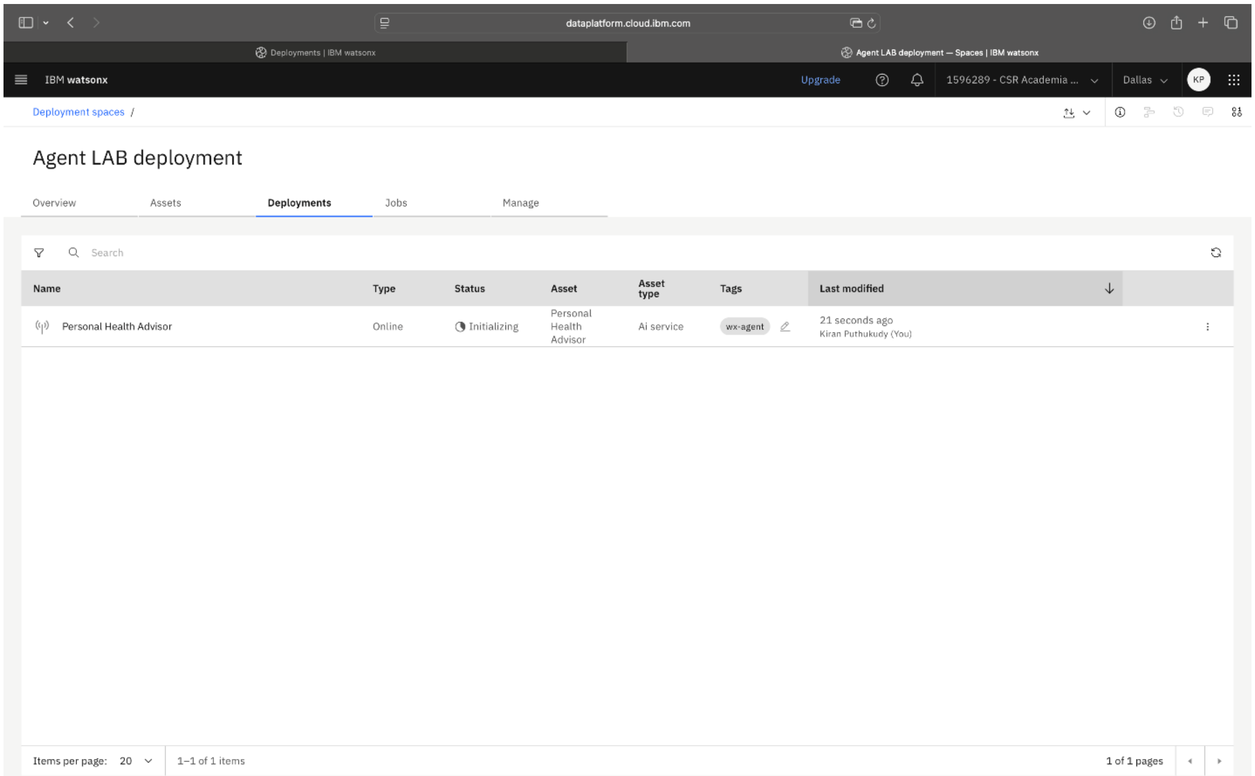
Alt text: An image showing the **Deploy as an AI service** section with the **Target deployment space** drop-down menu displaying the selected option **Agent LAB deployment**, along with the **Deploy** option

7. A status update appears stating "Deployment has started". Select **View status**.



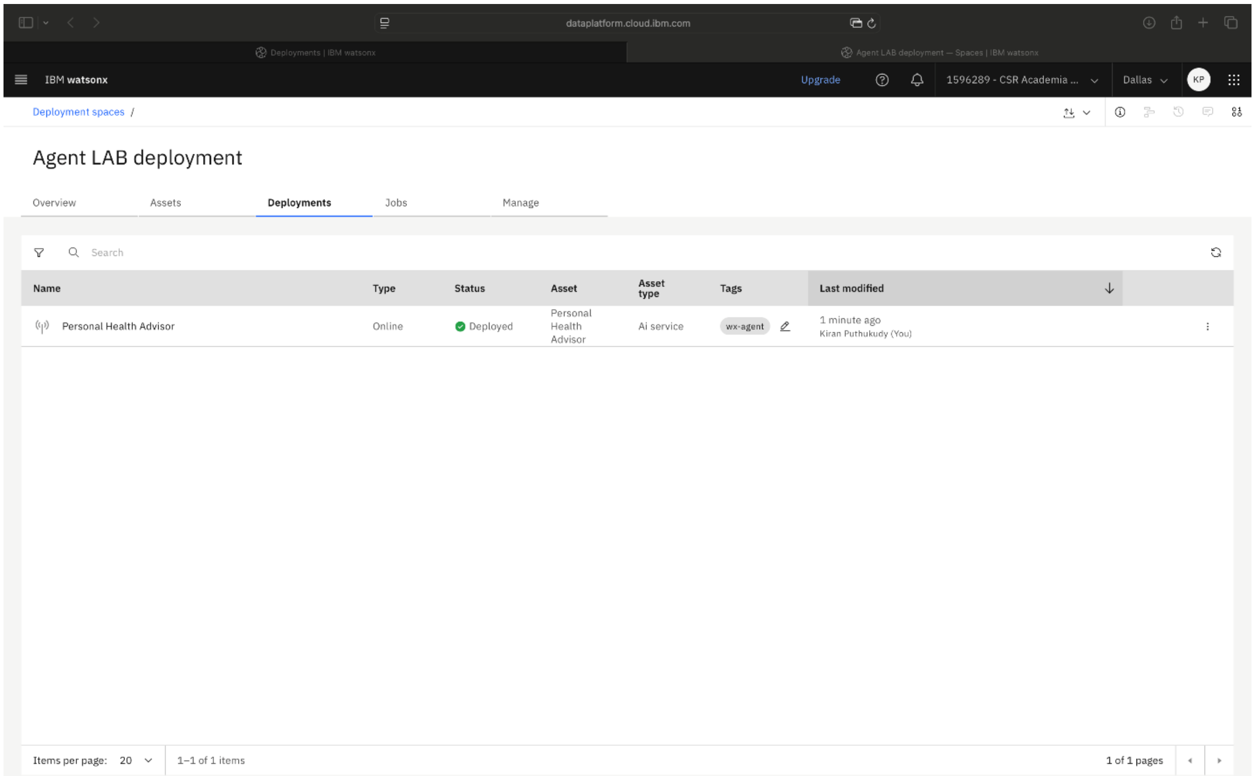
Alt text: An image showing the watsonx Agent interface with the status update “Deployment has started”

8. The system status displays “Initializing”. Wait for the initialization to complete or reload to start again. Select **Next** to continue.



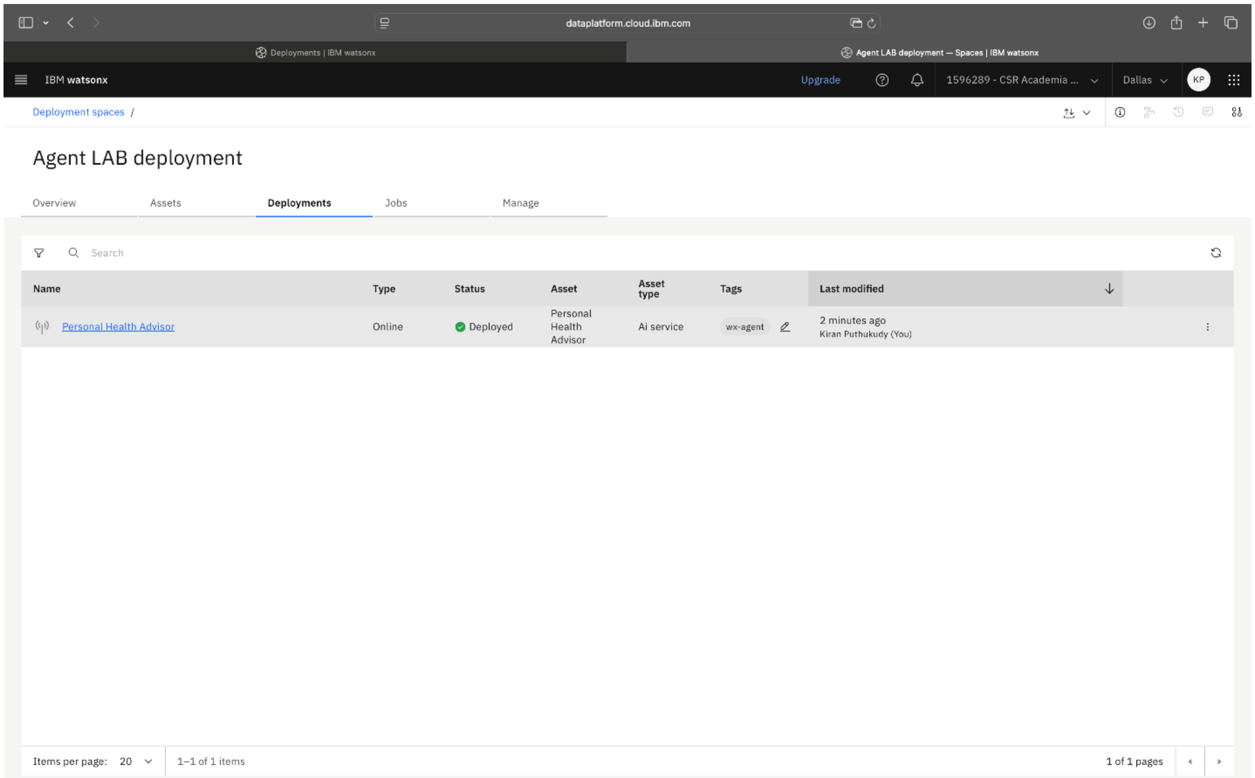
Alt text: An image showing the IBM watson Agent LAB deployment screen, showing the Personal Health Advisor deployment with the "Initializing" status and tagged as **wx-agent**

9. When ready, the status updates to "Deployed" and the type displays as "Online". Select **Next** to continue.



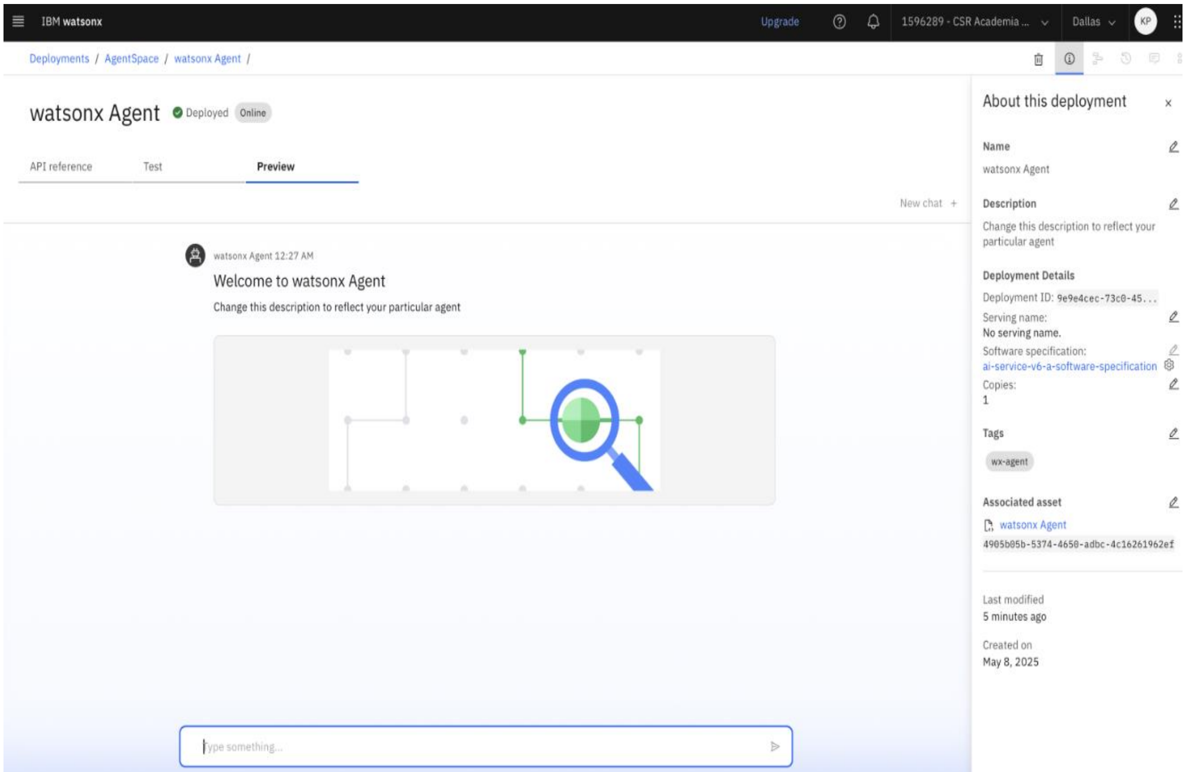
Alt text: An image showing the IBM watson Agent LAB deployment screen, showing a Personal Health Advisor deployment with the "Deployed" status and tagged as **wx-agent**

10. Select **Personal Health Advisor** to access the deployed agent.



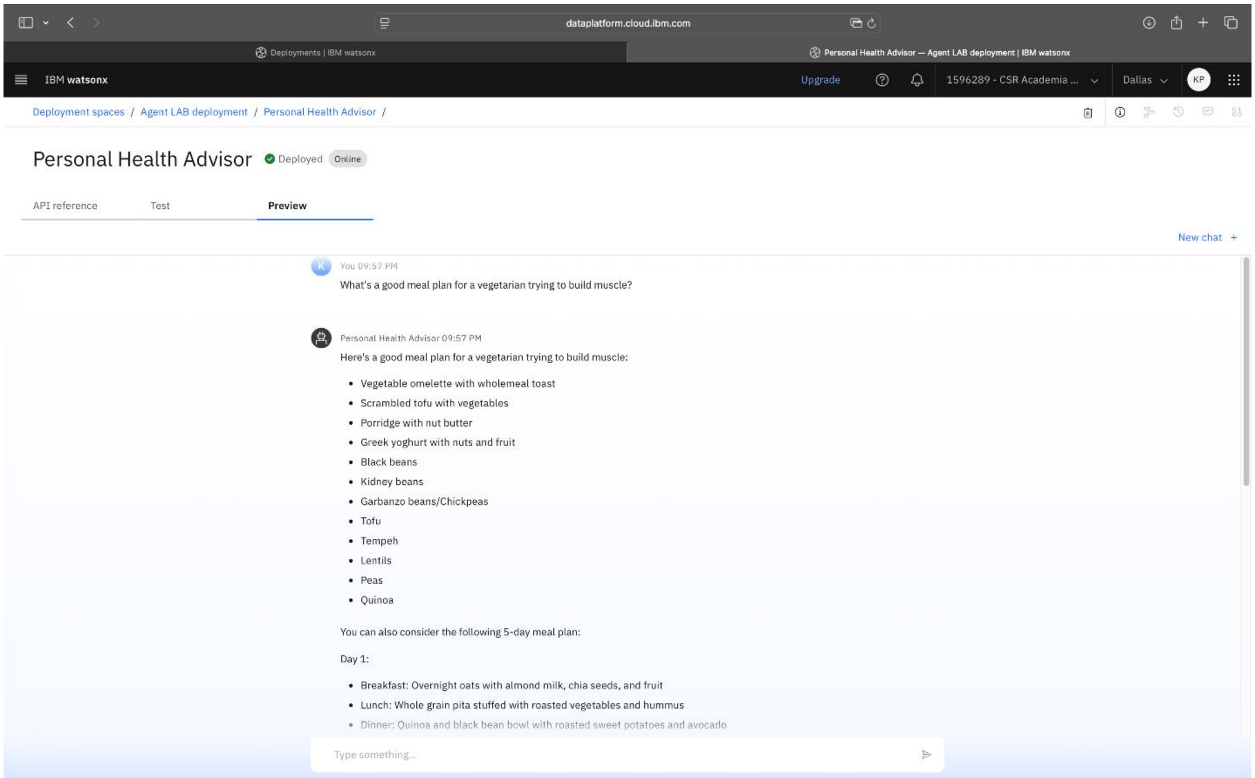
Alt text: An image showing the IBM watson Agent LAB deployment screen, showing a Personal Health Advisor deployment with the "Deployed" status and tagged as **wx-agent**

11. Select **Preview**.



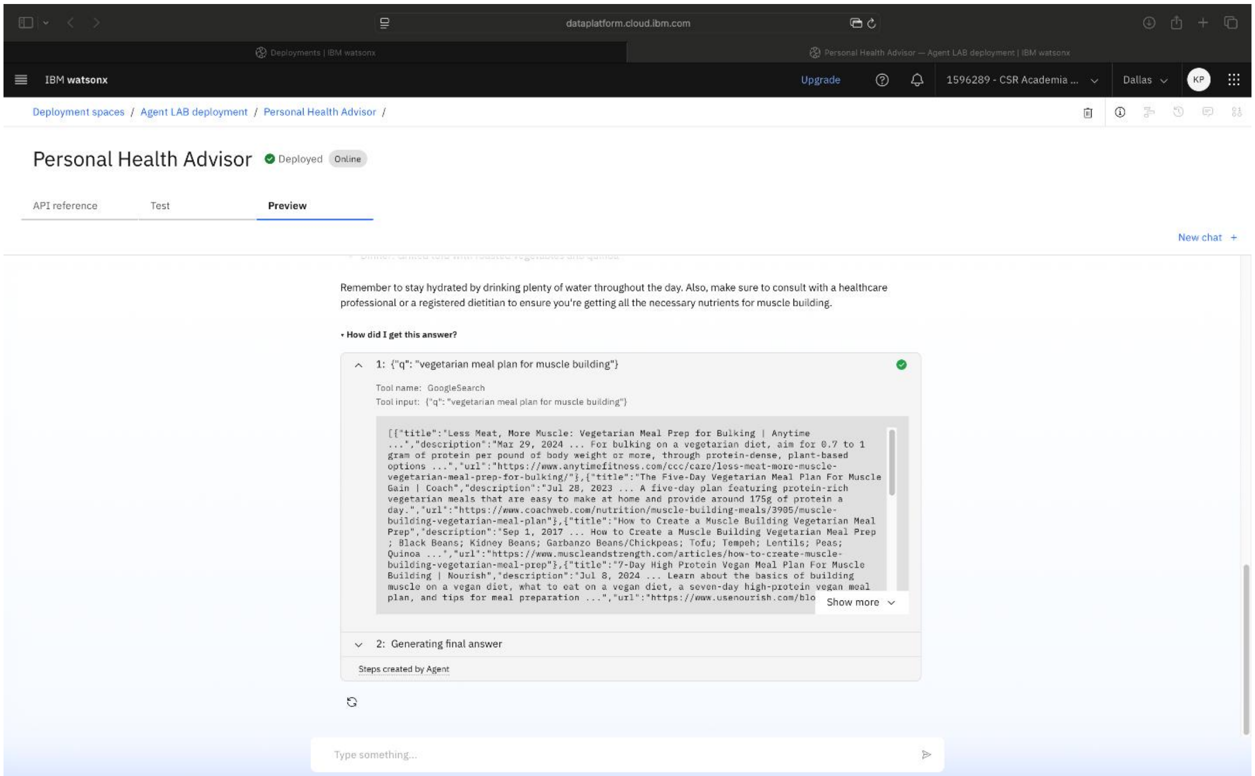
Alt text: An image showing the watsonx Agent interface with a welcome message displayed in the **Preview** section

12. Test the agent using prompts such as “What’s a good meal plan for a vegetarian trying to build muscle?” or “It’s raining today. Suggest an indoor workout.”



Alt text: An image showing a conversation between the user and the watsonx Agent about a meal plan for vegetarians to build muscle

13. Review the agent's reasoning path and the final response. Select **Next** to continue.



Alt text: An image showing how the agent retrieved a vegetarian meal plan for muscle building

Conclusion

Congratulations! You've successfully built and deployed a Personal Health Advisor agent using watsonx Agent Lab. This lab demonstrated how to define a custom prompt, use integrated tools, and deploy your agent in a live environment. You are now equipped to design, deploy, and iterate intelligent agents for real-world use cases.